

RTL Design Sherpa

RTL AMBA AXI5-Lite Module Reference — Rev 0.90

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1 AXI5-Lite (AXIL5) Modules

Location: `rtl/amba/axil5/` **Sibling family:** AXI4-Lite

1.1 Overview

Sixteen modules mirroring the AXI4-Lite family one for one, with the AXI5-Lite transaction ID and optional signal groups threaded through. Same channel set, same handshakes, same single-beat semantics – what AXI5-Lite adds is the ID and sideband (IHI 0022 Issue L, B2.1.5 and Table B2.2).

If you came here looking for a full AXI5-Lite protocol stack, read the next section before you instantiate anything.

1.2 Functional Description

These modules **transport** AXI5-Lite signals. They do not execute AXI5-Lite semantics:

- The ID (AWID/BID, ARID/RID) is carried in both directions; reflecting AxID onto BID/RID is the Subordinate's job (A5.1.1), and responses to different IDs may be reordered by it (B2.1.5).
- USER and TRACE are carried end to end unmodified.
- POISON is carried. It is never generated and never checked.

Those behaviours belong to the endpoints on either side. Read this before treating the family as a full AXI5-Lite protocol stack – it is not one.

AXI4-Lite vs AXI5-Lite, as this library implements them:

Feature	AXI4-Lite	AXI5-Lite
Channels	AW, W, B, AR, R	same
Burst support	none	none
Data width	32 or 64 (IHI 0022H B1.1.2)	any AXI width, 8..1024 (L Table A3.19; Table B2.4 adds no Lite limit)
Transaction ID	none	`AWID`/`BID`, `ARID`/`RID`, `ID_WIDTH` bits (L Table A5.2: 0..32). Optional on a

Feature	AXI4-Lite	AXI5-Lite
		Manager, mandatory on a Subordinate (Table B2.2, YS), so `ID_WIDTH` may be 0 on the `axil5_master_*` modules and must be >= 1 on the `axil5_slave_*` ones
User sideband	not present	`AxUSER`/`WUSER`/`BUSER`/`RUSER`, gated by `ENABLE_USER`
Trace	not present	`AxTRACE`/`BTRACE`/`RTRACE`, gated by `ENABLE_TRACE`
Poison	not present	`WPOISON`/`RPOISON`, one bit per 64 data bits, gated by `ENABLE_POISON`
Exclusive access	not present	not present either (B2.1.5); no `AxLOCK`
MPAM / MECID / NSAID / Loopback	not present	not present either (Table B2.2 “N”)

Both data width and `ID_WIDTH` are checked at elaboration by every module’s `param_check` block; an illegal value stops the build.

With no ID and every group disabled the two are the same interface. On the Manager-side modules (`ID_WIDTH = 0`, every `ENABLE_*` = 0) the packed payload widths match channel for channel, which is what lets one testbench bind to either family – `val/amba/test_axil5_axil4_equivalence.py` pins it.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP were carried behind `ENABLE_*` parameters until issue #102; Table B2.2 marks them not present on AXI5-Lite, so they were removed, and the ID the table requires was added.

1.3 Usage Example

Two configurations – a Manager-side module configured down to AXI4-Lite, and a Subordinate-side module with its ID and sideband:

```
// AXI4-Lite-equivalent Manager side: no ID, every group off --
// payload widths identical to axil4_master_rd
```

```
axil5_master_rd #(
    .AXIL_ADDR_WIDTH (32),
    .AXIL_DATA_WIDTH (32),
    .ID_WIDTH        (0),
    .ENABLE_USER (0), .ENABLE_TRACE (0), .ENABLE_POISON (0)
) u_plain ( /* ... */ );

// Subordinate side with IDs (mandatory here) and the trace/user
sideband
axil5_slave_rd #(
    .AXIL_ADDR_WIDTH (32),
    .AXIL_DATA_WIDTH (64),
    .ID_WIDTH        (6),
    .USER_WIDTH      (8),
    .ENABLE_USER (1), .ENABLE_TRACE (1), .ENABLE_POISON (1)
) u_sub ( /* ... */ );
```

1.4 Design Notes

Known limitations:

- No protocol checking of the ID or the optional groups. The monitored variants observe handshakes, addresses, responses and timing; since amba #104 their monitor also sees AxID and BID/RID, so responses a Subordinate reorders across IDs are paired with their own transactions (W in AW order). `axi_monitor_filtered` has no ports for USER, TRACE or POISON.
- No poison generation or checking.
- Timing diagrams are not yet drawn for this family.

1.5 Related Modules

The family, grouped by what each variant adds:

1.5.1 Transport (4)

Module	Channels
axil5_master_rd	AR, R
axil5_master_wr	AW, W, B
axil5_slave_rd	AR, R
axil5_slave_wr	AW, W, B

1.5.2 Clock-gated (4)

Module	Adds
axil5_master_rd_cg	One `amba_clock_gate_ctrl` over the whole inner module
axil5_master_wr_cg	One `amba_clock_gate_ctrl` over the whole inner module
axil5_slave_rd_cg	One `amba_clock_gate_ctrl` over the whole inner module
axil5_slave_wr_cg	One `amba_clock_gate_ctrl` over the whole inner module

1.5.3 Monitored (4)

Module	Adds
axil5_master_rd_mon	`axi_monitor_filtered`, monbus packet output
axil5_master_wr_mon	`axi_monitor_filtered`, monbus packet output
axil5_slave_rd_mon	`axi_monitor_filtered`, monbus packet output
axil5_slave_wr_mon	`axi_monitor_filtered`, monbus packet output

1.5.4 Monitored and clock-gated (4)

Module	Adds
axil5_master_rd_mon_cg	Both, with ready held low while gated
axil5_master_wr_mon_cg	Both, with ready held low while gated
axil5_slave_rd_mon_cg	Both, with ready held low while gated
axil5_slave_wr_mon_cg	Both, with ready held low while gated

1.6 Testing

`val/amba/test_axil5_*.py` – sixteen runners, 42 parameterised cases, all passing. They drive the modules with **every optional group enabled**, which makes them the only tests in the repo that exercise the AXI5-Lite sideband against real transport RTL rather than a behavioural model.

The testbench classes in `bin/TBClasses/axil5/` are the AXI4-Lite ones with the component factories swapped and the group widths set in `COMPONENT_KWARGS`. One definition of the traffic and its checks.

Also here: `test_axil5_opt_signals.py` drives `axil5_opt_slave`, a behavioural slave under `rtl/amba/axil5/test-modules/` built purely so the BFMs had a DUT carrying every optional port. It is test collateral – do not instantiate it in a design.

1.7 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR/R and AW/W/B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): the family carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19, ID width per Table A5.2.
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. “Theory of Latency-Insensitive Design.” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 – relay stations: storage on a back-pressured channel adds latency without changing function.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.

- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- Chipressian. *Crack the Hardware Interview – from RTL Designers' Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Bruno, F., Eschemann, G. *The FPGA Programming Handbook*, 2nd ed., Packt, 2024, ch. 8 “Introduction to AXI”, §8.3 “AXI4 interfaces (full and AXI-Lite)” – textbook treatment of the AXI-Lite channel set (secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`); clock-gating papers in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

1.8 Navigation

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2 AXIL5 Master Read

Module: `axil5_master_rd.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_rd`

2.1 Overview

The AXIL5 Master Read provides a buffered AXI5-Lite read interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_rd` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled

2.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID

Parameter	Type	Default	Description
			(Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AR	int	2	
SKID_DEPTH_R	int	4	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
ARSize	int	AW + 3 +	
RSize	int	DW + 2 +	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

2.3 Ports

Your logic drives the fub_* side; the m_axil_* side faces the bus.

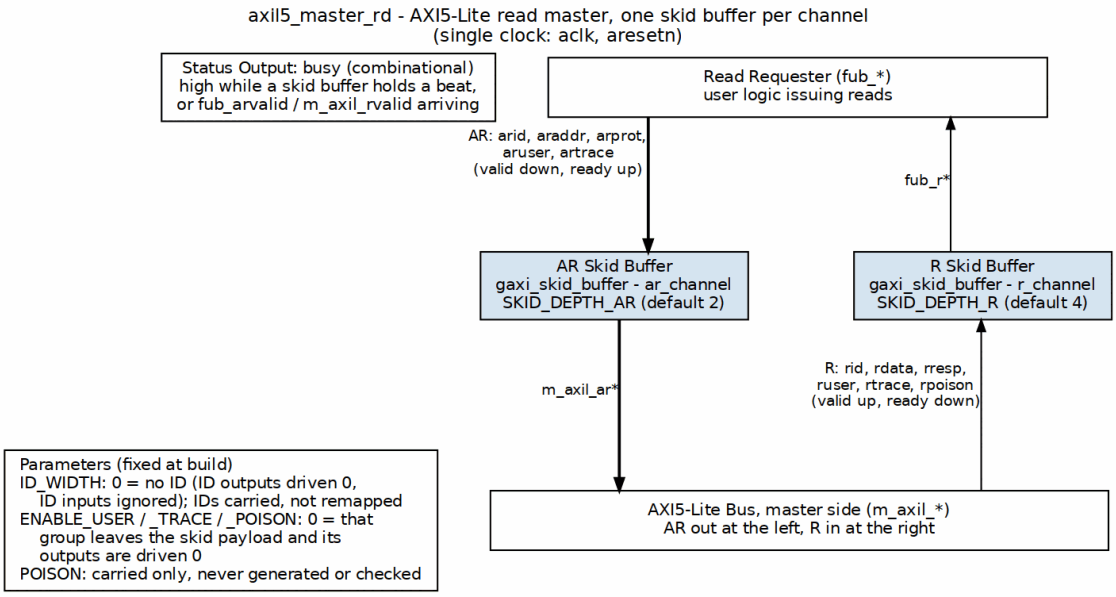
Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	

Port	Direction	Width	Description
fub_arid	Input	[IW-1:0]	
fub_araddr	Input	[AW-1:0]	
fub_arprot	Input	[2:0]	
fub_aruser	Input	[UW-1:0]	
fub_artrace	Input	1	
fub_arvalid	Input	1	
fub_arready	Output	1	
m_axil_arid	Output	[IW-1:0]	
m_axil_araddr	Output	[AW-1:0]	
m_axil_arprot	Output	[2:0]	
m_axil_aruser	Output	[UW-1:0]	
m_axil_artrace	Output	1	
m_axil_arvalid	Output	1	
m_axil_arready	Input	1	
m_axil_rid	Input	[IW-1:0]	
m_axil_rdata	Input	[DW-1:0]	
m_axil_rresp	Input	[1:0]	
m_axil_ruser	Input	[UW-1:0]	
m_axil_rtrace	Input	1	
m_axil_rpoison	Input	[PW-1:0]	
m_axil_rvalid	Input	1	
m_axil_rready	Output	1	
fub_rid	Output	[IW-1:0]	
fub_rdata	Output	[DW-1:0]	
fub_rresp	Output	[1:0]	
fub_ruser	Output	[UW-1:0]	
fub_rtrace	Output	1	
fub_rpoison	Output	[PW-1:0]	

Port	Direction	Width	Description
fub_rvalid	Output	1	
fub_rready	Input	1	
busy	Output	1	

2.4 Functional Description

2.4.1 Block diagram



axil5_master_rd block diagram

The requester (`fub_*`) sits at the top and the AXI5-Lite bus (`m_axil_*`) is the bar at the bottom. AR runs down through its `gaxi_skid_buffer` to the bottom, and the R response climbs back up its own lane on the right through a second skid buffer, so every channel crosses exactly one buffer. This is the axil4 structure; AXI5-Lite only adds carried fields – the ID on every channel and the optional USER, TRACE and POISON groups, each dropped from the skid payload (outputs driven 0) when its `ENABLE_*` parameter is 0, as the Parameters legend states. `busy` is high while any skid buffer holds a beat or a valid is arriving (fub_arvalid, m_axil_rvalid), the same definition as every other AMBA base module.

2.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

2.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

2.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the

Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With `ID_WIDTH = 0` (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

2.5 Timing Characteristics

2.5.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AR`	2 entries	Skid depth on the AR channel
`SKID_DEPTH_R`	4 entries	Skid depth on the R channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

2.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

2.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_rd #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4)
) u_axil5_master_rd (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // fub_ar
    .fub_arid            (fub_arid),
    // ... `fub_araddr`, `fub_arprot`, `fub_aruser`, +3 more
    // m_axil_ar
    .m_axil_arid         (m_axil_arid),
    // ... `m_axil_araddr`, `m_axil_arprot`, `m_axil_aruser`, +3 more
    // m_axil_r
    .m_axil_rid          (m_axil_rid),
    // ... `m_axil_rdata`, `m_axil_rresp`, `m_axil_ruser`, +4 more
    // fub_r
    .fub_rid             (fub_rid),
    // ... `fub_rdata`, `fub_rresp`, `fub_ruser`, +4 more
    // status
    .busy                (busy)
);
```

2.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS`` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/`DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

2.8 Related Modules

- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- axil5_master_wr_mon
- `axil4_master_rd` – the AXI4-Lite counterpart
- AXI4-Lite modules

2.9 Testing

`val/amba/test_axil5_master_rd.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

2.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. “Theory of Latency-Insensitive Design.” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 – relay stations: storage on a back-pressured channel adds latency without changing function.
- Chipressian. *Crack the Hardware Interview – from RTL Designers’ Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Bruno, F., Eschemann, G. *The FPGA Programming Handbook*, 2nd ed., Packt, 2024, ch. 8 “Introduction to AXI”, §8.3 “AXI4 interfaces (full and AXI-Lite)” – textbook treatment of the AXI-Lite channel set (secondary source).

- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

2.11 Navigation

AXI5-Lite index | AMBA overview

3 AXIL5 Master Read, Clock-Gated

Module: `axil5_master_rd_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_rd_cg`

3.1 Overview

The AXIL5 Master Read, Clock-Gated provides a buffered AXI5-Lite read interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_rd_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled
- One `amba_clock_gate_ctrl` gating the whole inner module

3.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and

Parameter	Type	Default	Description
			ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AR	int	2	
SKID_DEPTH_R	int	4	
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

3.3 Ports

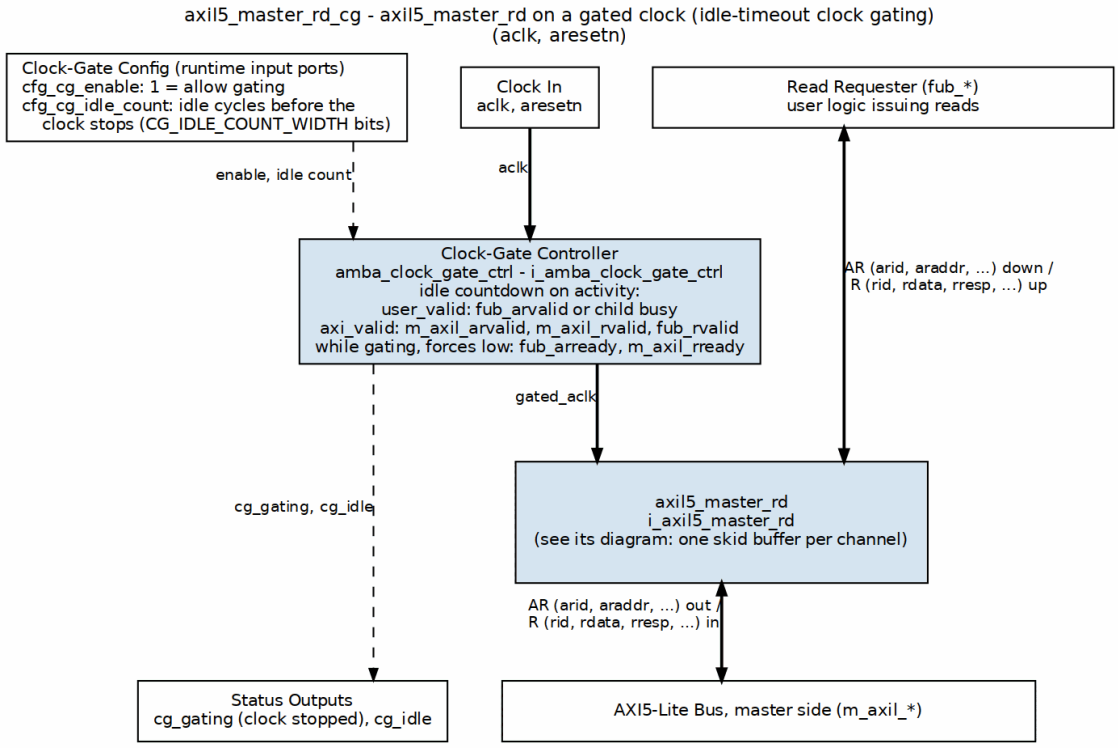
Your logic drives the fub_* side; the m_axil_* side faces the bus. The cfg_cg_* inputs run the clock gate.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
cfg_cg_enable	Input	1	
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	
fub_arid	Input	[IW-1:0]	
fub_araddr	Input	[AW-1:0]	
fub_arprot	Input	[2:0]	
fub_aruser	Input	[UW-1:0]	
fub_artrace	Input	1	
fub_arvalid	Input	1	
fub_arready	Output	1	
m_axil_arid	Output	[IW-1:0]	
m_axil_araddr	Output	[AW-1:0]	
m_axil_arprot	Output	[2:0]	
m_axil_aruser	Output	[UW-1:0]	
m_axil_artrace	Output	1	
m_axil_arvalid	Output	1	
m_axil_arready	Input	1	
m_axil_rid	Input	[IW-1:0]	
m_axil_rdata	Input	[DW-1:0]	
m_axil_rresp	Input	[1:0]	
m_axil_ruser	Input	[UW-1:0]	
m_axil_rtrace	Input	1	
m_axil_rpoison	Input	[PW-1:0]	
m_axil_rvalid	Input	1	
m_axil_rready	Output	1	
fub_rid	Output	[IW-1:0]	
fub_rdata	Output	[DW-1:0]	

Port	Direction	Width	Description
fub_rresp	Output	[1:0]	
fub_ruser	Output	[UW-1:0]	
fub_rtrace	Output	1	
fub_rpoison	Output	[PW-1:0]	
fub_rvalid	Output	1	
fub_rready	Input	1	
cg_gating	Output	1	Active gating indicator
cg_idle	Output	1	All buffers empty indicator

3.4 Functional Description

3.4.1 Block diagram



axil5_master_rd_cg block diagram

axil5_master_rd is one child box (see its own diagram), clocked by gated_aclk. The heavy clock column runs from aclk through amba_clock_gate_ctrl, which counts down an idle window and stops the clock when nothing is active; its label lists the exact user_valid and axi_valid activity terms and the readies it forces low while gating, so no transfer is accepted on a stopped clock. cg_gating and cg_idle leave as status outputs.

3.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by ID_WIDTH); each group is gated by its own ENABLE_* parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

3.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

3.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

3.5 Timing Characteristics

3.5.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AR	2 entries	Skid depth on the AR channel
SKID_DEPTH_R	4 entries	Skid depth on the R channel

Each channel traverses one gaxi_skid_buffer. That module registers both rd_valid and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

3.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

3.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_rd_cg #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4),
    .CG_IDLE_COUNT_WIDTH (4)
) u_axil5_master_rd_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // configuration
    .cfg_cg_enable       (cfg_cg_enable),
    .cfg_cg_idle_count   (cfg_cg_idle_count),
    // fub_ar
    .fub_arid            (fub_arid),
    // ... `fub_araddr`, `fub_arprot`, `fub_aruser`, +3 more
    // m_axil_ar
    .m_axil_arid         (m_axil_arid),
    // ... `m_axil_araddr`, `m_axil_arprot`, `m_axil_aruser`, +3 more
    // m_axil_r
    .m_axil_rid          (m_axil_rid),
    // ... `m_axil_rdata`, `m_axil_rresp`, `m_axil_ruser`, +4 more
    // fub_r
    .fub_rid             (fub_rid),
    // ... `fub_rdata`, `fub_rresp`, `fub_ruser`, +4 more
    // status
```

```
    .cg_gating          (cg_gating),
    .cg_idle            (cg_idle)
);
```

3.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS `` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/ `DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

A peer's READY must never enter the activity term. A consumer that parks its response-ready high while idle is behaving correctly; folding that signal into `user_valid` pins this block permanently awake and defeats gating entirely, silently, because function is unaffected. This wrapper wakes on VALIDs and pending work only. `val/amba/test_cg_peer_ready.py` parks the peer READY high, holds every VALID low, and requires `cg_gating` – see `vault/handbook/design/clock-gating-activity-terms.md`.

`busy` is not brought out here. The plain `_cg` wrappers consume the base module's `busy` internally as a wake term; use `cg_idle` or `cg_gating` for idle detection. The `_mon_cg` variants do expose `busy`.

`cfg_cg_enable` is not a kill switch. It arms gating and reaches `amba_clock_gate_ctrl` only. With it low the clock free-runs and the module behaves exactly like its base.

3.8 Related Modules

- axil5_master_rd
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- axil5_master_wr_mon
- `axil4_master_rd_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

3.9 Testing

`val/amba/test_axil5_master_rd_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

3.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).

- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

3.11 Navigation

AXI5-Lite index | AMBA overview

4 AXIL5 Master Read Monitor

Module: `axil5_master_rd_mon.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_rd_mon`

4.1 Overview

The AXIL5 Master Read Monitor provides a buffered AXI5-Lite read interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_rd_mon` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled
- Integrated transaction monitor emitting monbus packets

4.2 Parameters

A lite-monitor version of this wrapper exists as `axil5_master_rd_monlite`: the same core and taps with `axi_monitor_lite` instead of `axi_monitor_filtered`, about a fifth of the monitor gates, error/timeout/completion/threshold packets only. Selected in generated bridges by `mon_preset = "lite"`.

Parameter	Type	Default	Description
`SKID_DEPTH_AR`	int	2	
`SKID_DEPTH_R`	int	4	
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and

Parameter	Type	Default	Description
			a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_MONITOR	bit	1'b1	0 = omit monitor, tie outputs
N_ADDR_RANGES	int	0	0 = address-range checker disabled
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
ID_FILTER_ENABLE	bit	1'b0	Per-instance ID-slice filter, inherited from the shared monitor core. Since amba #104 the wrapper feeds the AXI5-Lite ID, so the slice selects on AxID as on the AXI4 wrappers.
ADDR_FILTER_ENABLE	bit	1'b0	
ID_MATCH_BASE	int	0	First ID this

Parameter	Type	Default	Description
			instance owns when ID_FILTER_ENABLE =1. Compared against the AXI5-Lite AxID.
ID_MATCH_COUNT	int	0	Number of IDs owned from ID_MATCH_BASE. 0 = all IDs (the filter passes everything).
ACTIVE_TRANS_THRESHOLD	int	MAX_TRANSACTIONS / 2	
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.

Parameter	Type	Default	Description
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h000A	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h01	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit

Parameter	Type	Default	Description
			Unit ID for monitor packets)

The derived parameters (`AW`, `DW`, `UW`, … and the `*Size` payload widths) are computed from the ones above. **Do not override them.** Forcing a `*Size` to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how `test_axil5_master_wr` failed until `d6266344` removed those overrides.

4.3 Ports

Your logic drives the `fub_axil_*` side; the `m_axil_*` side faces the bus. The `cfg_*` and `monbus_*` ports belong to the monitor.

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cam_clear`	Input	1	sync clear of the monitor trans CAM
`fub_axil_arid`	Input	`[IW-1:0]`	
`fub_axil_araddr`	Input	`[AW-1:0]`	
`fub_axil_arprot`	Input	`[2:0]`	
`fub_axil_aruser`	Input	`[UW-1:0]`	
`fub_axil_artrace`	Input	1	
`fub_axil_arvalid`	Input	1	
`fub_axil_arready`	Output	1	
`fub_axil_rid`	Output	`[IW-1:0]`	
`fub_axil_rdata`	Output	`[DW-1:0]`	
`fub_axil_rresp`	Output	`[1:0]`	
`fub_axil_ruser`	Output	`[UW-1:0]`	
`fub_axil_rtrace`	Output	1	
`fub_axil_rpoison`	Output	`[PW-1:0]`	
`fub_axil_rvalid`	Output	1	

Port	Direction	Width	Description
fub_axil_rready	Input	1	
m_axil_arid	Output	[IW-1:0]	
m_axil_araddr	Output	[AW-1:0]	
m_axil_arprot	Output	[2:0]	
m_axil_aruser	Output	[UW-1:0]	
m_axil_artrace	Output	1	
m_axil_arvalid	Output	1	
m_axil_arready	Input	1	
m_axil_rid	Input	[IW-1:0]	
m_axil_rdata	Input	[DW-1:0]	
m_axil_rresp	Input	[1:0]	
m_axil_ruser	Input	[UW-1:0]	
m_axil_rtrace	Input	1	
m_axil_rpoison	Input	[PW-1:0]	
m_axil_rvalid	Input	1	
m_axil_rready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets

Port	Direction	Width	Description
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_addr_check_enable	Input	1	

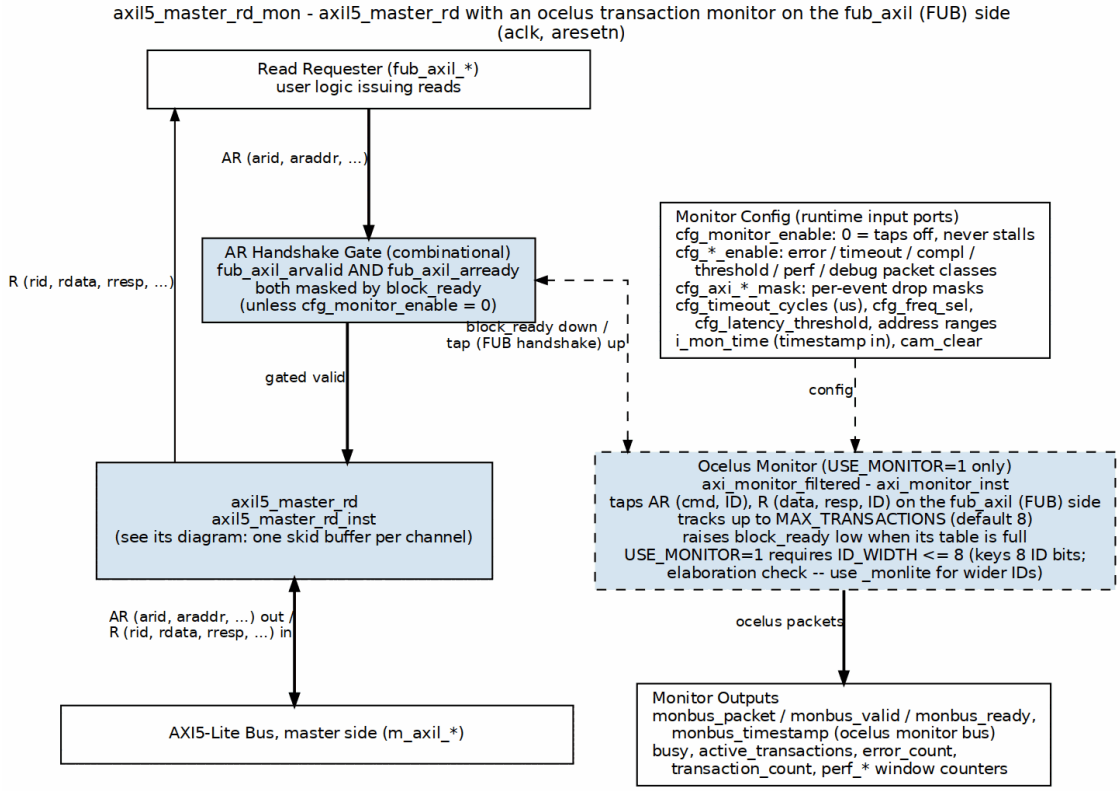
Port	Direction	Width	Description
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES >

Port	Direction	Width	Description
			0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
debug_block_ready	Output	1	
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	

Port	Direction	Width	Description
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

4.4 Functional Description

4.4.1 Block diagram



axil5_master_rd_mon block diagram

`axil5_master_rd` is one child box (see its own diagram). The AR request first passes a handshake gate that masks valid and ready with `block_ready`, so a full monitor table stalls new requests instead of losing them. The optional ocelus monitor (`axi_monitor_filtered`, dashed because `USE_MONITOR=0` removes it) taps the FUB-side (`fub_axil`) handshake – the same point the handshake gate acts on, so there is no skid lag – including the transaction IDs, and emits monitor-bus packets as the heavy edge. With `USE_MONITOR=1` the build requires `ID_WIDTH <= 8`; the `_monlite` wrapper takes wider IDs.

4.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1
POISON	`ENABLE_POISON`	`WPOISON`, `RPOISON`	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

4.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, `ID_WIDTH=0`, groups off	AXI5-Lite defaults (`ID_WIDTH=4`, groups on)
`ARSize`	35	35	44
`RSize`	34	34	44
`AWSize`	35	35	44
`WSize`	36	36	41
`BSize`	2	2	11

(at `AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32` and the default widths. `ID_WIDTH = 0` is legal only on the Manager-side `axil5_master_*` modules; a Subordinate must carry the ID, so the `axil5_slave_*` modules require `ID_WIDTH >= 1` and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what `val/amba/test_axil5_axil4_equivalence.py` pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both `axil4_master_rd` and `axil5_master_rd` elaborated with `ID_WIDTH=0` and every `ENABLE_*` = 0.

4.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With `ID_WIDTH = 0` (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

4.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

4.6 Timing Characteristics

4.6.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AR	2 entries	Skid depth on the AR channel
SKID_DEPTH_R	4 entries	Skid depth on the R channel

Each channel traverses one gaxi_skid_buffer. That module registers both rd_valid and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

4.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: ARSize, AWSize, WSize, RSize and BSize are conditional sums over the ENABLE_* parameters, so disabling a group narrows the storage without changing latency.

4.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_rd_mon #(
    .SKID_DEPTH_AR      (2),
    .SKID_DEPTH_R       (4),
    .AXIL_ADDR_WIDTH    (32),
    .AXIL_DATA_WIDTH    (32),
    .ACLK_MHZ           (100),
    .USE_MONITOR        (1'b1),
    .N_ADDR_RANGES      (0),
    .UNIT_ID            (8'h01),
    .AGENT_ID           (16'h000A),
```

```
    .MAX_TRANSACTIONS    (8),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .ID_FILTER_ENABLE    (1'b0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ID_MATCH_BASE       (0),
    .ID_MATCH_COUNT      (0),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_master_rd_mon (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_araddr`, `fub_axil_arprot`, `fub_axil_arlock`,
+31 more
    // m_axil_ar
    .m_axil_arid         (m_axil_arid),
    // ... `m_axil_araddr`, `m_axil_arprot`, `m_axil_aruser`, +3 more
    // m_axil_r
    .m_axil_rid          (m_axil_rid),
    // ... `m_axil_rdata`, `m_axil_rresp`, `m_axil_ruser`, +4 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +28 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

4.8 Related Modules

- axil5_master_rd

- axil5_master_rd_cg
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- axil5_master_wr_mon
- `axil4_master_rd_mon` – the AXI4-Lite counterpart
- AXI4-Lite modules

4.9 Testing

`val/amba/test_axil5_master_rd_mon.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

4.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an

AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.

- Cortadella, J., Kishinevsky, M., Grundmann, B. "Synthesis of Synchronous Elastic Architectures." *Proc. 43rd Design Automation Conf. (DAC '06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel's `gaxi_skid_buffer`.
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Chipressian. *Crack the Hardware Interview – from RTL Designers' Perspective: Architecture and Micro-architecture Design*, ch. 2 "Handshake Protocols", Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

4.11 Navigation

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5 AXIL5 Master Read Monitor, Clock-Gated

Module: `axil5_master_rd_mon_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_rd_mon_cg`

5.1 Overview

The AXIL5 Master Read Monitor, Clock-Gated provides a buffered AXI5-Lite read interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_rd_mon_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled
- Integrated transaction monitor emitting monbus packets
- Monitor plus clock gating; ready signals held low while gated

5.2 Parameters

A lite-monitor version of this clock-gated wrapper exists as `axil5_master_rd_monlite_cg`: the same gating logic around `axil5_master_rd_monlite` instead of `axil5_master_rd_mon`, about a fifth of the monitor gates.

Parameter	Type	Default	Description
`SKID_DEPTH_AR`	int	2	
`SKID_DEPTH_R`	int	4	
`AXIL_ADDR_WIDTH`	int	32	

Parameter	Type	Default	Description
AXIL_DATA_WIDTH	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
USE_MONITOR	bit	1'b1	0 = omit monitor in inner mon; outputs tied
N_ADDR_RANGES	int	0	0 = address-range checker disabled
ADDR_FILTER_ENABLE	bit	1'b0	0 = address filter inert
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOG	bit	1'b1	

Parameter	Type	Default	Description
IC			
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h000A	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to

Parameter	Type	Default	Description
			identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h01	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit Unit ID for monitor packets)

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

5.3 Ports

Your logic drives the fub_axil_* side; the m_axil_* side faces the bus. The cfg_* and monbus_* ports belong to the monitor, and cfg_cg_* runs the clock gate.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
cam_clear	Input	1	sync clear of the monitor trans CAM
fub_axil_arid	Input	[IW-1:0]	

Port	Direction	Width	Description
fub_axil_araddr	Input	[AW-1:0]	
fub_axil_arprot	Input	[2:0]	
fub_axil_aruser	Input	[UW-1:0]	
fub_axil_artrace	Input	1	
fub_axil_arvalid	Input	1	
fub_axil_arready	Output	1	
fub_axil_rid	Output	[IW-1:0]	
fub_axil_rdata	Output	[DW-1:0]	
fub_axil_rresp	Output	[1:0]	
fub_axil_ruser	Output	[UW-1:0]	
fub_axil_rtrace	Output	1	
fub_axil_rpoison	Output	[PW-1:0]	
fub_axil_rvalid	Output	1	
fub_axil_rready	Input	1	
m_axil_arid	Output	[IW-1:0]	
m_axil_araddr	Output	[AW-1:0]	
m_axil_arprot	Output	[2:0]	
m_axil_aruser	Output	[UW-1:0]	
m_axil_artrace	Output	1	
m_axil_arvalid	Output	1	
m_axil_arready	Input	1	
m_axil_rid	Input	[IW-1:0]	
m_axil_rdata	Input	[DW-1:0]	
m_axil_rresp	Input	[1:0]	
m_axil_ruser	Input	[UW-1:0]	
m_axil_rtrace	Input	1	
m_axil_rpoison	Input	[PW-1:0]	
m_axil_rvalid	Input	1	
m_axil_rready	Output	1	
cfg_monitor_enable	Input	1	Enable

Port	Direction	Width	Description
			monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_ma	Input	[15:0]	Individual

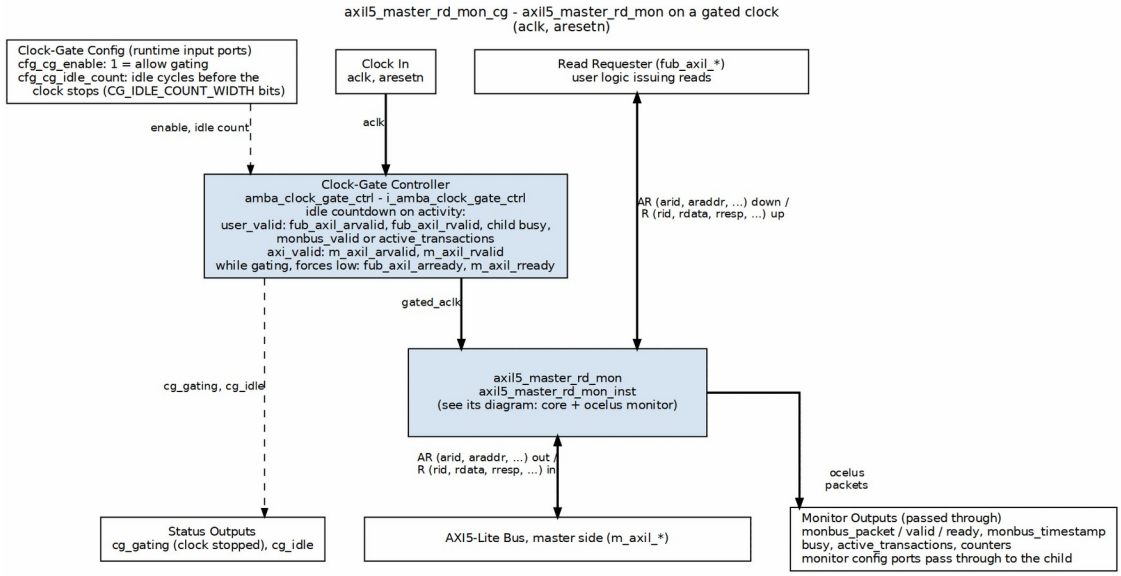
Port	Direction	Width	Description
sk			completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_cg_enable	Input	1	Enable clock gating
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	Idle cycles before gating
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when

Port	Direction	Width	Description
			monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
cg_gating	Output	1	Gated clock is

Port	Direction	Width	Description
			stopped
cg_idle	Output	1	No activity observed
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

5.4 Functional Description

5.4.1 Block diagram



axil5_master_rd_mon_cg block diagram

`axil5_master_rd_mon` (core plus ocelus monitor) is one child box (see its own diagram), clocked by `gated_aclk` from `amba_clock_gate_ctrl`. The controller's label lists the activity terms, which add `monbus_valid` and `active_transactions` so the clock keeps running while the monitor still has packets or open transactions. Monitor-bus packets and the monitor configuration pass straight through to the child.

5.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1

Group	Parameter	Signals on this module	Width
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

5.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

5.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them “not present” on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group’s output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

5.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

5.6 Timing Characteristics

5.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AR`	2 entries	Skid depth on the AR channel
`SKID_DEPTH_R`	4 entries	Skid depth on the R channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the**

unstalled case – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

5.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

5.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_rd_mon_cg #(
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ACLK_MHZ            (100),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .UNIT_ID             (8'h01),
    .AGENT_ID            (16'h000A),
    .MAX_TRANSACTIONS    (8),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .CG_IDLE_COUNT_WIDTH (4),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
```

```
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_master_rd_mon_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_araddr`, `fub_axil_arprot`, `fub_axil_arlock`,
+32 more
    // m_axil_ar
    .m_axil_arid         (m_axil_arid),
    // ... `m_axil_araddr`, `m_axil_arprot`, `m_axil_aruser`, +3 more
    // m_axil_r
    .m_axil_rid          (m_axil_rid),
    // ... `m_axil_rdata`, `m_axil_rresp`, `m_axil_ruser`, +4 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +30 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

5.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_wr
- axil5_master_wr_cg
- axil5_master_wr_mon
- `axil4_master_rd_mon_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

5.9 Testing

`val/amba/test_axil5_master_rd_mon_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in

COMPONENT_KWARGS to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (bin/TBClasses/axil5/axil5_id_check.py); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

5.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (ID_WIDTH) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the ID_WIDTH=0, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after cfg_cg_idle_count idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under clock_gate_ctrl.
- READY held LOW while gated is legal: a receiver may keep READY LOW until it chooses to accept (IHI 0022H §A3.2.1; IHI 0022L §A2.3.1). No paper (standard technique).

- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

5.11 Navigation

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6 AXIL5 Master Write

Module: `axil5_master_wr.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_wr`

6.1 Overview

The AXIL5 Master Write provides a buffered AXI5-Lite write interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_wr` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled

6.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID

Parameter	Type	Default	Description
			(Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AW	int	2	
SKID_DEPTH_W	int	4	
SKID_DEPTH_B	int	2	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
SW	int	DW / 8	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AWSize	int	AW + 3 +	
WSize	int	DW + SW +	
BSize	int	2 +	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

6.3 Ports

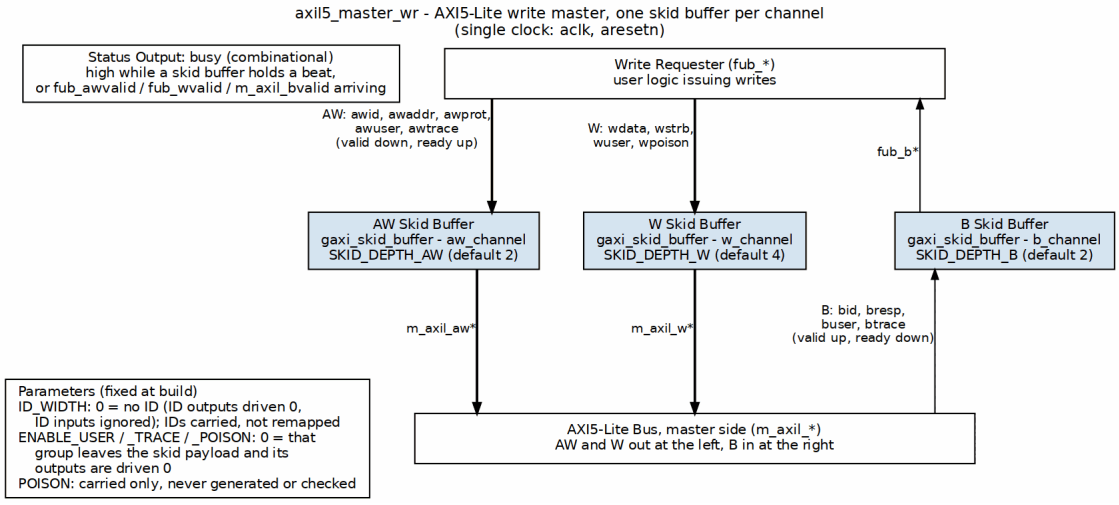
Your logic drives the fub_* side; the m_axil_* side faces the bus.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
fub_awid	Input	[IW-1:0]	
fub_awaddr	Input	[AW-1:0]	
fub_awprot	Input	[2:0]	
fub_awuser	Input	[UW-1:0]	
fub_awtrace	Input	1	
fub_awvalid	Input	1	
fub_awready	Output	1	
m_axil_awid	Output	[IW-1:0]	
m_axil_awaddr	Output	[AW-1:0]	
m_axil_awprot	Output	[2:0]	
m_axil_awuser	Output	[UW-1:0]	
m_axil_awtrace	Output	1	
m_axil_awvalid	Output	1	
m_axil_awready	Input	1	
fub_wdata	Input	[DW-1:0]	
fub_wstrb	Input	[SW-1:0]	
fub_wuser	Input	[UW-1:0]	
fub_wpoison	Input	[PW-1:0]	
fub_wvalid	Input	1	
fub_wready	Output	1	
m_axil_wdata	Output	[DW-1:0]	
m_axil_wstrb	Output	[SW-1:0]	
m_axil_wuser	Output	[UW-1:0]	
m_axil_wpoison	Output	[PW-1:0]	
m_axil_wvalid	Output	1	
m_axil_wready	Input	1	

Port	Direction	Width	Description
m_axil_bid	Input	[IW-1:0]	
m_axil_bresp	Input	[1:0]	
m_axil_buser	Input	[UW-1:0]	
m_axil_btrace	Input	1	
m_axil_bvalid	Input	1	
m_axil_bready	Output	1	
fub_bid	Output	[IW-1:0]	
fub_bresp	Output	[1:0]	
fub_buser	Output	[UW-1:0]	
fub_btrace	Output	1	
fub_bvalid	Output	1	
fub_bready	Input	1	
busy	Output	1	

6.4 Functional Description

6.4.1 Block diagram



axil5_master_wr block diagram

The requester (`fub_*`) sits at the top and the AXI5-Lite bus (`m_axil_*`) is the bar at the bottom. AW and W run down through their own `gaxi_skid_buffers`, and the B response climbs back up its own lane on the right through a third, so every channel crosses exactly one buffer. This is the

axil4 structure; AXI5-Lite only adds carried fields – the ID on every channel and the optional USER, TRACE and POISON groups, each dropped from the skid payload (outputs driven 0) when its ENABLE_* parameter is 0, as the Parameters legend states. busy is high while any skid buffer holds a beat or a valid is arriving (fub_awvalid, fub_wvalid, m_axil_bvalid), the same definition as every other AMBA base module.

6.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by ID_WIDTH); each group is gated by its own ENABLE_* parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

6.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

6.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

6.5 Timing Characteristics

6.5.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AW	2 entries	Skid depth on the AW channel
SKID_DEPTH_W	4 entries	Skid depth on the W channel
SKID_DEPTH_B	2 entries	Skid depth on the B channel

Each channel traverses one gaxi_skid_buffer. That module registers both rd_valid and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the

depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

6.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

6.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_wr #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (4),
    .SKID_DEPTH_B        (2)
) u_axil5_master_wr (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // fub_aw
    .fub_awid            (fub_awid),
    // ... `fub_awaddr`, `fub_awprot`, `fub_awuser`, +3 more
    // m_axil_aw
    .m_axil_awid         (m_axil_awid),
    // ... `m_axil_awaddr`, `m_axil_awprot`, `m_axil_awuser`, +3 more
    // fub_w
    .fub_wdata           (fub_wdata),
    // ... `fub_wstrb`, `fub_wuser`, `fub_wpoison`, +2 more
    // m_axil_w
    .m_axil_wdata        (m_axil_wdata),
    // ... `m_axil_wstrb`, `m_axil_wuser`, `m_axil_wpoison`, +2 more
    // m_axil_b
```

```
    .m_axil_bid           (m_axil_bid),
    // ... `m_axil_bresp`, `m_axil_buser`, `m_axil_btrace`, +2 more
    // fub_b
    .fub_bid              (fub_bid),
    // ... `fub_bresp`, `fub_buser`, `fub_btrace`, +2 more
    // status
    .busy                 (busy)
);
```

6.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS `` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/ `DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads “Extracting 4 bits from only 2 bit number”, and it is how this family’s test port broke.

The ID is AXI5-Lite’s, not the monitor’s. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

6.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon

- axil5_master_rd_mon_cg
- axil5_master_wr_cg
- axil5_master_wr_mon
- `axil4_master_wr` – the AXI4-Lite counterpart
- AXI4-Lite modules

6.9 Testing

`val/amba/test_axil5_master_wr.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

6.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*,

pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel's `gaxi_skid_buffer`.

- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. "Theory of Latency-Insensitive Design." *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 – relay stations: storage on a back-pressured channel adds latency without changing function.
- Chipressian. *Crack the Hardware Interview – from RTL Designers' Perspective: Architecture and Micro-architecture Design*, ch. 2 "Handshake Protocols", Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Bruno, F., Eschemann, G. *The FPGA Programming Handbook*, 2nd ed., Packt, 2024, ch. 8 "Introduction to AXI", §8.3 "AXI4 interfaces (full and AXI-Lite)" – textbook treatment of the AXI-Lite channel set (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

6.11 Navigation

AXI5-Lite index | AMBA overview

7 AXIL5 Master Write, Clock-Gated

Module: `axil5_master_wr_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_wr_cg`

7.1 Overview

The AXIL5 Master Write, Clock-Gated provides a buffered AXI5-Lite write interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_wr_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled
- One `amba_clock_gate_ctrl` gating the whole inner module

7.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and

Parameter	Type	Default	Description
			ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AW	int	2	
SKID_DEPTH_W	int	4	
SKID_DEPTH_B	int	2	
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
SW	int	DW / 8	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

7.3 Ports

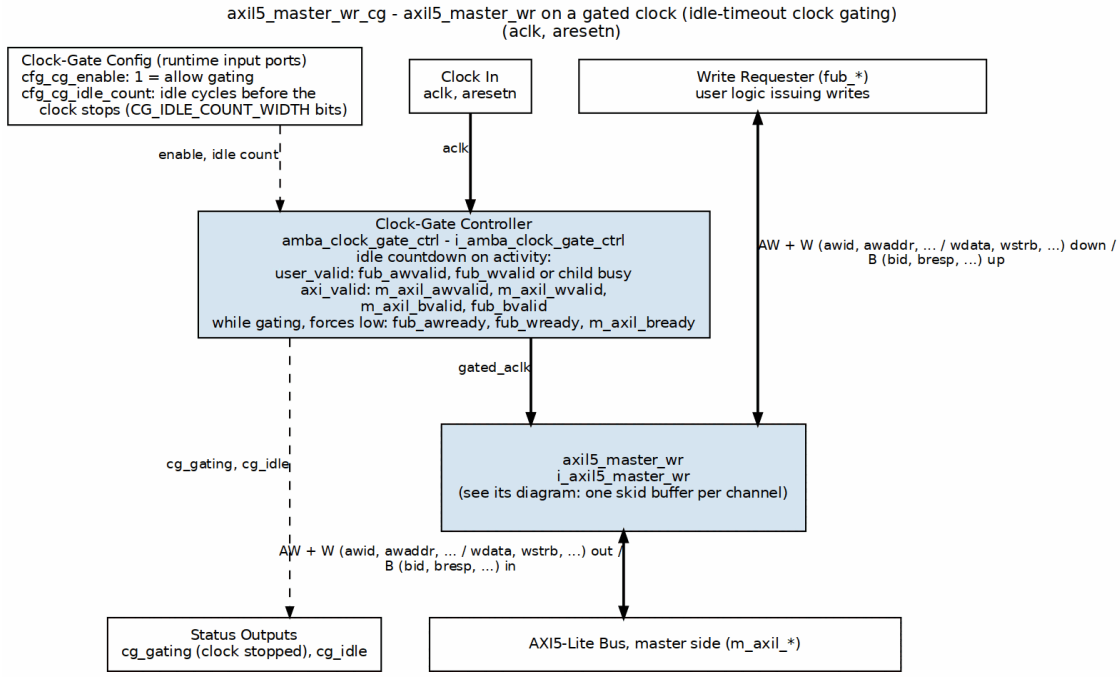
Your logic drives the `fub_*` side; the `m_axil_*` side faces the bus. The `cfg_cg_*` inputs run the clock gate.

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cfg_cg_enable`	Input	1	
`cfg_cg_idle_count`	Input	`[CG_IDLE_COUNT_WIDTH-1:0]`	
`fub_awid`	Input	`[IW-1:0]`	
`fub_awaddr`	Input	`[AW-1:0]`	
`fub_awprot`	Input	`[2:0]`	
`fub_awuser`	Input	`[UW-1:0]`	
`fub_awtrace`	Input	1	
`fub_awvalid`	Input	1	
`fub_awready`	Output	1	
`m_axil_awid`	Output	`[IW-1:0]`	
`m_axil_awaddr`	Output	`[AW-1:0]`	
`m_axil_awprot`	Output	`[2:0]`	
`m_axil_awuser`	Output	`[UW-1:0]`	
`m_axil_awtrace`	Output	1	
`m_axil_awvalid`	Output	1	
`m_axil_awready`	Input	1	
`fub_wdata`	Input	`[DW-1:0]`	
`fub_wstrb`	Input	`[SW-1:0]`	
`fub_wuser`	Input	`[UW-1:0]`	
`fub_wpoison`	Input	`[PW-1:0]`	
`fub_wvalid`	Input	1	
`fub_wready`	Output	1	
`m_axil_wdata`	Output	`[DW-1:0]`	

Port	Direction	Width	Description
m_axil_wstrb	Output	[SW-1:0]	
m_axil_wuser	Output	[UW-1:0]	
m_axil_wpoison	Output	[PW-1:0]	
m_axil_wvalid	Output	1	
m_axil_wready	Input	1	
m_axil_bid	Input	[IW-1:0]	
m_axil_bresp	Input	[1:0]	
m_axil_buser	Input	[UW-1:0]	
m_axil_btrace	Input	1	
m_axil_bvalid	Input	1	
m_axil_bready	Output	1	
fub_bid	Output	[IW-1:0]	
fub_bresp	Output	[1:0]	
fub_buser	Output	[UW-1:0]	
fub_btrace	Output	1	
fub_bvalid	Output	1	
fub_bready	Input	1	
cg_gating	Output	1	Active gating indicator
cg_idle	Output	1	All buffers empty indicator

7.4 Functional Description

7.4.1 Block diagram



axil5_master_wr_cg block diagram

`axil5_master_wr` is one child box (see its own diagram), clocked by `gated_aclk`. The heavy clock column runs from `aclk` through `amba_clock_gate_ctrl`, which counts down an idle window and stops the clock when nothing is active; its label lists the exact `user_valid` and `axi_valid` activity terms and the readies it forces low while gating, so no transfer is accepted on a stopped clock. `cg_gating` and `cg_idle` leave as status outputs.

7.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`

Group	Parameter	Signals on this module	Width
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

7.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

7.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them “not present” on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group’s output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

7.5 Timing Characteristics

7.5.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	4 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

7.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

7.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_wr_cg #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (4),
    .SKID_DEPTH_B        (2),
    .CG_IDLE_COUNT_WIDTH (4)
) u_axil5_master_wr_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // configuration
    .cfg_cg_enable       (cfg_cg_enable),
    .cfg_cg_idle_count   (cfg_cg_idle_count),
    // fub_aw
    .fub_awid            (fub_awid),
    // ... `fub_awaddr`, `fub_awprot`, `fub_awuser`, +3 more
    // m_axil_aw
    .m_axil_awid         (m_axil_awid),
    // ... `m_axil_awaddr`, `m_axil_awprot`, `m_axil_awuser`, +3 more
    // fub_w
    .fub_wdata           (fub_wdata),
    // ... `fub_wstrb`, `fub_wuser`, `fub_wpoison`, +2 more
    // m_axil_w
    .m_axil_wdata        (m_axil_wdata),
    // ... `m_axil_wstrb`, `m_axil_wuser`, `m_axil_wpoison`, +2 more
    // m_axil_b
    .m_axil_bid          (m_axil_bid),
    // ... `m_axil_bresp`, `m_axil_buser`, `m_axil_btrace`, +2 more
    // fub_b
    .fub_bid             (fub_bid),
    // ... `fub_bresp`, `fub_buser`, `fub_btrace`, +2 more
    // status
    .cg_gating           (cg_gating),
    .cg_idle             (cg_idle)
);
```

7.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager

side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS`` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/`DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

A peer's READY must never enter the activity term. A consumer that parks its response-ready high while idle is behaving correctly; folding that signal into `user_valid` pins this block permanently awake and defeats gating entirely, silently, because function is unaffected. This wrapper wakes on VALIDs and pending work only. `val/amba/test_cg_peer_ready.py` parks the peer READY high, holds every VALID low, and requires `cg_gating` – see `vault/handbook/design/clock-gating-activity-terms.md`.

`busy` is not brought out here. The plain `_cg` wrappers consume the base module's `busy` internally as a wake term; use `cg_idle` or `cg_gating` for idle detection. The `_mon_cg` variants do expose `busy`.

`cfg_cg_enable` is not a kill switch. It arms gating and reaches `amba_clock_gate_ctrl` only. With it low the clock free-runs and the module behaves exactly like its base.

7.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr

- axil5_master_wr_mon
- `axil4_master_wr_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

7.9 Testing

`val/amba/test_axil5_master_wr_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

7.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.

- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

7.11 Navigation

AXI5-Lite index | AMBA overview

8 AXIL5 Master Write Monitor

Module: axil5_master_wr_mon.sv **Location:** rtl/amba/axil5/ **AXI4-Lite counterpart:** axil4_master_wr_mon

8.1 Overview

The AXIL5 Master Write Monitor provides a buffered AXI5-Lite write interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally axil4_master_wr_mon with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal AXIL_DATA_WIDTH or ID_WIDTH
- Transaction ID (AWID/BID, ARID/RID; ID_WIDTH) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with ID_WIDTH=0 and all groups disabled
- Integrated transaction monitor emitting monbus packets

8.2 Parameters

A lite-monitor version of this wrapper exists as axil5_master_wr_monlite: the same core and taps with axi_monitor_lite instead of axi_monitor_filtered, about a fifth of the monitor gates, error/timeout/completion/threshold packets only. Selected in generated bridges by mon_preset = "lite".

Parameter	Type	Default	Description
SKID_DEPTH_AW	int	2	
SKID_DEPTH_W	int	2	
SKID_DEPTH_B	int	2	
AXIL_ADDR_WIDTH	int	32	
AXIL_DATA_WIDTH	int	32	Any AXI data

Parameter	Type	Default	Description
			width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_MONITOR	bit	1'b1	0 = omit monitor, tie outputs
N_ADDR_RANGES	int	0	0 = address-range checker disabled
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
ID_FILTER_ENABLE	bit	1'b0	Per-instance ID-slice filter, inherited from the shared monitor core. Since amba #104 the wrapper feeds the AXI5-Lite ID, so the slice selects on AxID as on the AXI4 wrappers.
ADDR_FILTER_ENABLE	bit	1'b0	

Parameter	Type	Default	Description
ID_MATCH_BASE	int	0	First ID this instance owns when ID_FILTER_ENABLE =1. Compared against the AXI5-Lite AxID.
ID_MATCH_COUNT	int	0	Number of IDs owned from ID_MATCH_BASE. 0 = all IDs (the filter passes everything).
ACTIVE_TRANS_THRESHOLD	int	MAX_TRANSACTIONS / 2	
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L

Parameter	Type	Default	Description
			Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h000B	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h01	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the

Parameter	Type	Default	Description
			collector. (8-bit Unit ID for monitor packets)

The derived parameters (`AW`, `DW`, `UW`, … and the `*Size` payload widths) are computed from the ones above. **Do not override them.** Forcing a `*Size` to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how `test_axil5_master_wr` failed until `d6266344` removed those overrides.

8.3 Ports

Your logic drives the `fub_axil_*` side; the `m_axil_*` side faces the bus. The `cfg_*` and `monbus_*` ports belong to the monitor.

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cam_clear`	Input	1	sync clear of the monitor trans CAM
`fub_axil_awid`	Input	`[IW-1:0]`	
`fub_axil_awaddr`	Input	`[AW-1:0]`	
`fub_axil_awprot`	Input	`[2:0]`	
`fub_axil_awuser`	Input	`[UW-1:0]`	
`fub_axil_awtrace`	Input	1	
`fub_axil_awvalid`	Input	1	
`fub_axil_awready`	Output	1	
`fub_axil_wdata`	Input	`[DW-1:0]`	
`fub_axil_wstrb`	Input	`[DW/8-1:0]`	
`fub_axil_wuser`	Input	`[UW-1:0]`	
`fub_axil_wpoison`	Input	`[PW-1:0]`	
`fub_axil_wvalid`	Input	1	
`fub_axil_wready`	Output	1	

Port	Direction	Width	Description
fub_axil_bid	Output	[IW-1:0]	
fub_axil_bresp	Output	[1:0]	
fub_axil_buser	Output	[UW-1:0]	
fub_axil_btrace	Output	1	
fub_axil_bvalid	Output	1	
fub_axil_bready	Input	1	
m_axil_awid	Output	[IW-1:0]	
m_axil_awaddr	Output	[AW-1:0]	
m_axil_awprot	Output	[2:0]	
m_axil_awuser	Output	[UW-1:0]	
m_axil_awtrace	Output	1	
m_axil_awvalid	Output	1	
m_axil_awready	Input	1	
m_axil_wdata	Output	[DW-1:0]	
m_axil_wstrb	Output	[DW/8-1:0]	
m_axil_wuser	Output	[UW-1:0]	
m_axil_wpoison	Output	[PW-1:0]	
m_axil_wvalid	Output	1	
m_axil_wready	Input	1	
m_axil_bid	Input	[IW-1:0]	
m_axil_bresp	Input	[1:0]	
m_axil_buser	Input	[UW-1:0]	
m_axil_btrace	Input	1	
m_axil_bvalid	Input	1	
m_axil_bready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection

Port	Direction	Width	Description
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask

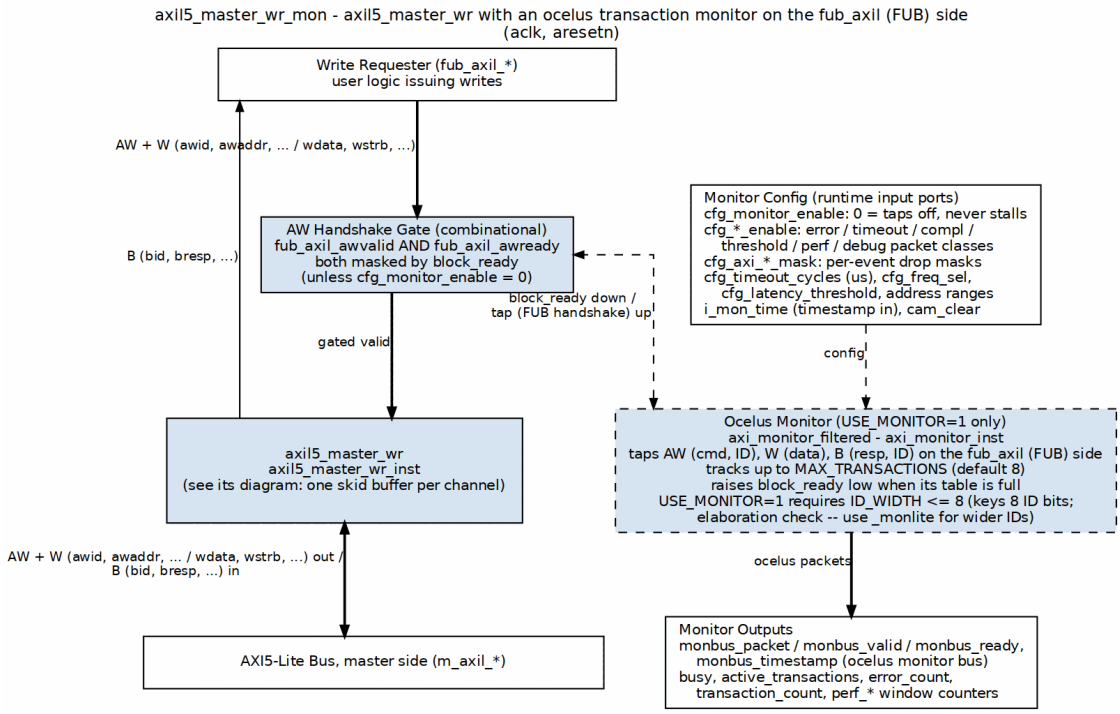
Port	Direction	Width	Description
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the

Port	Direction	Width	Description
			group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
debug_block_ready	Output	1	
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force	Input	1	

Port	Direction	Width	Description
_close			
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

8.4 Functional Description

8.4.1 Block diagram



axil5_master_wr_mon block diagram

axil5_master_wr is one child box (see its own diagram). The AW request first passes a handshake gate that masks valid and ready with block_ready, so a full monitor table stalls new requests instead of losing them. The optional ocelus monitor (axi_monitor_filtered, dashed because USE_MONITOR=0 removes it) taps the FUB-side (fub_axil) handshake – the same point the handshake gate acts on, so there is no skid lag – including the transaction IDs, and emits monitor-bus packets as the heavy edge. With USE_MONITOR=1 the build requires ID_WIDTH <= 8; the _monlite wrapper takes wider IDs.

8.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by ID_WIDTH); each group is gated by its own ENABLE_* parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

8.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

8.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

8.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to axi_monitor_filtered, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so ID_WIDTH > 8 is refused at elaboration here (the _monlite wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

ACLK_MHZ is not decoration. It builds the microsecond tick LUT in counter_freq_invariant. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

NUM_BANKS > 1 on a WRITE monitor requires USE_WDATA_ORDER_Q = 1. axi_monitor_trans_mgr fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

8.6 Timing Characteristics

8.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	2 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

8.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

8.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_wr_mon #(
    .SKID_DEPTH_AW      (2),
    .SKID_DEPTH_W       (2),
```

```
    .SKID_DEPTH_B        (2),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ACLK_MHZ            (100),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .UNIT_ID             (8'h01),
    .AGENT_ID            (16'h000B),
    .MAX_TRANSACTIONS    (8),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .ID_FILTER_ENABLE    (1'b0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ID_MATCH_BASE       (0),
    .ID_MATCH_COUNT      (0),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_master_wr_mon (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_awaddr`, `fub_axil_awprot`, `fub_axil_awlock`,
+35 more
    // m_axil_aw
    .m_axil_awid         (m_axil_awid),
    // ... `m_axil_awaddr`, `m_axil_awprot`, `m_axil_awuser`, +3 more
    // m_axil_w
    .m_axil_wdata        (m_axil_wdata),
    // ... `m_axil_wstrb`, `m_axil_wuser`, `m_axil_wpoison`, +2 more
    // m_axil_b
    .m_axil_bid          (m_axil_bid),
    // ... `m_axil_bresp`, `m_axil_buser`, `m_axil_btrace`, +2 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
```

```
`cfg_perf_enable`, +28 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

8.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_master_wr_mon` – the AXI4-Lite counterpart
- AXI4-Lite modules

8.9 Testing

`val/amba/test_axil5_master_wr_mon.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

8.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID

and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).

- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Chipressian. *Crack the Hardware Interview – from RTL Designers’ Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

8.11 Navigation

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9 AXIL5 Master Write Monitor, Clock-Gated

Module: `axil5_master_wr_mon_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_master_wr_mon_cg`

9.1 Overview

The AXIL5 Master Write Monitor, Clock-Gated provides a buffered AXI5-Lite write interface for master devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_master_wr_mon_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- Bit-identical to the AXI4-Lite module of the same name with `ID_WIDTH=0` and all groups disabled
- Integrated transaction monitor emitting monbus packets
- Monitor plus clock gating; ready signals held low while gated

9.2 Parameters

A lite-monitor version of this clock-gated wrapper exists as `axil5_master_wr_monlite_cg`: the same gating logic around `axil5_master_wr_monlite` instead of `axil5_master_wr_mon`, about a fifth of the monitor gates.

Parameter	Type	Default	Description
`SKID_DEPTH_AW`	int	2	
`SKID_DEPTH_W`	int	2	
`SKID_DEPTH_B`	int	2	

Parameter	Type	Default	Description
AXIL_ADDR_WIDTH	int	32	
AXIL_DATA_WIDTH	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
USE_MONITOR	bit	1'b1	0 = omit monitor in inner mon; outputs tied
N_ADDR_RANGES	int	0	0 = address-range checker disabled
ADDR_FILTER_ENABLE	bit	1'b0	0 = address filter inert
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	

Parameter	Type	Default	Description
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 0..32; 0 = no ID (Manager side, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h000B	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to

Parameter	Type	Default	Description
			identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h01	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit Unit ID for monitor packets)

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

9.3 Ports

Your logic drives the fub_axil_* side; the m_axil_* side faces the bus. The cfg_* and monbus_* ports belong to the monitor, and cfg_cg_* runs the clock gate.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
cam_clear	Input	1	sync clear of the monitor trans CAM
fub_axil_awid	Input	[IW-1:0]	

Port	Direction	Width	Description
fub_axil_awaddr	Input	[AW-1:0]	
fub_axil_awprot	Input	[2:0]	
fub_axil_awuser	Input	[UW-1:0]	
fub_axil_awtrace	Input	1	
fub_axil_awvalid	Input	1	
fub_axil_awready	Output	1	
fub_axil_wdata	Input	[DW-1:0]	
fub_axil_wstrb	Input	[DW/8-1:0]	
fub_axil_wuser	Input	[UW-1:0]	
fub_axil_wpoison	Input	[PW-1:0]	
fub_axil_wvalid	Input	1	
fub_axil_wready	Output	1	
fub_axil_bid	Output	[IW-1:0]	
fub_axil_bresp	Output	[1:0]	
fub_axil_buser	Output	[UW-1:0]	
fub_axil_btrace	Output	1	
fub_axil_bvalid	Output	1	
fub_axil_bready	Input	1	
m_axil_awid	Output	[IW-1:0]	
m_axil_awaddr	Output	[AW-1:0]	
m_axil_awprot	Output	[2:0]	
m_axil_awuser	Output	[UW-1:0]	
m_axil_awtrace	Output	1	
m_axil_awvalid	Output	1	
m_axil_awready	Input	1	
m_axil_wdata	Output	[DW-1:0]	
m_axil_wstrb	Output	[DW/8-1:0]	
m_axil_wuser	Output	[UW-1:0]	
m_axil_wpoison	Output	[PW-1:0]	
m_axil_wvalid	Output	1	

Port	Direction	Width	Description
m_axil_wready	Input	1	
m_axil_bid	Input	[IW-1:0]	
m_axil_bresp	Input	[1:0]	
m_axil_buser	Input	[UW-1:0]	
m_axil_btrace	Input	1	
m_axil_bvalid	Input	1	
m_axil_bready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for

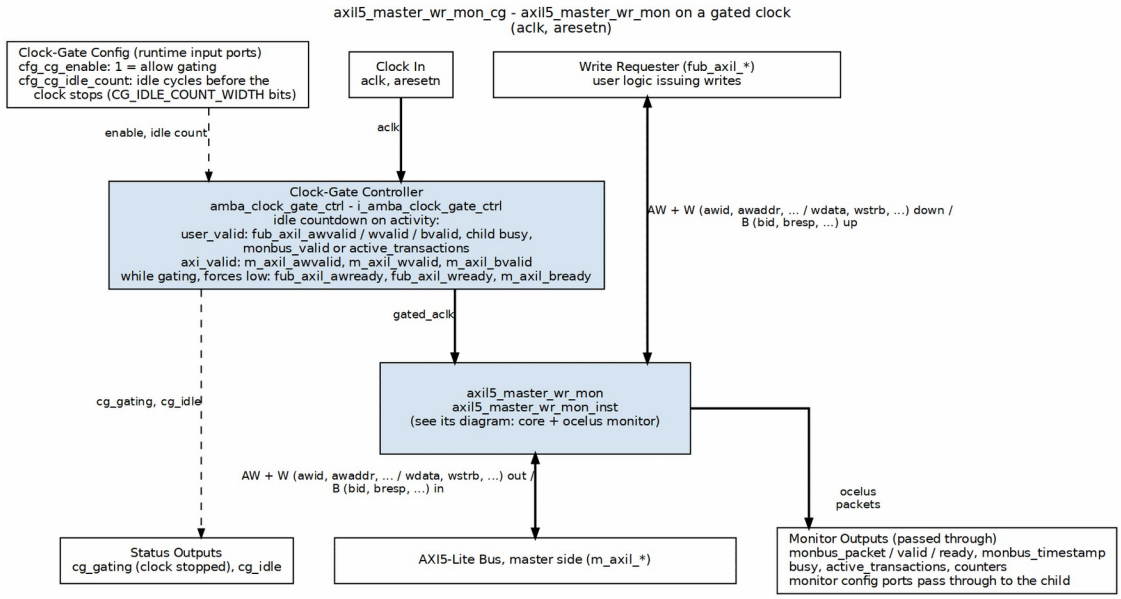
Port	Direction	Width	Description
			packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_cg_enable	Input	1	Enable clock gating
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	Idle cycles before gating
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	

Port	Direction	Width	Description
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_l

Port	Direction	Width	Description
			ow.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
cg_gating	Output	1	Gated clock is stopped
cg_idle	Output	1	No activity observed
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

9.4 Functional Description

9.4.1 Block diagram



axil5_master_wr_mon_cg block diagram

`axil5_master_wr_mon` (core plus ocelus monitor) is one child box (see its own diagram), clocked by `gated_aclk` from `amba_clock_gate_ctrl`. The controller's label lists the activity terms, which add `monbus_valid` and `active_transactions` so the clock keeps running while the monitor still has packets or open transactions. Monitor-bus packets and the monitor configuration pass straight through to the child.

9.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1

Group	Parameter	Signals on this module	Width
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

9.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

9.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them “not present” on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group’s output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

9.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

9.6 Timing Characteristics

9.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	2 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

9.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

9.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_master_wr_mon_cg #(
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (2),
    .SKID_DEPTH_B        (2),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ACLK_MHZ            (100),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .UNIT_ID             (8'h01),
    .AGENT_ID            (16'h000B),
    .MAX_TRANSACTIONS    (8),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .CG_IDLE_COUNT_WIDTH (4),
```

```
    .ID_WIDTH           (4),
    .USER_WIDTH         (4),
    .ENABLE_USER        (1),
    .ENABLE_TRACE       (1),
    .ENABLE_POISON      (1)
) u_axil5_master_wr_mon_cg (
    // clock/reset
    .aclk               (aclk),
    .aresetn            (aresetn),
    // status
    .cam_clear          (cam_clear),
    // ... `fub_axil_awaddr`, `fub_axil_awprot`, `fub_axil_awlock`,
+36 more
    // m_axil_aw
    .m_axil_awid        (m_axil_awid),
    // ... `m_axil_awaddr`, `m_axil_awprot`, `m_axil_awuser`, +3 more
    // m_axil_w
    .m_axil_wdata       (m_axil_wdata),
    // ... `m_axil_wstrb`, `m_axil_wuser`, `m_axil_wpoison`, +2 more
    // m_axil_b
    .m_axil_bid         (m_axil_bid),
    // ... `m_axil_bresp`, `m_axil_buser`, `m_axil_btrace`, +2 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +30 more
    // monitor bus
    .monbus_valid       (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

9.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_master_wr_mon_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

9.9 Testing

`val/amba/test_axil5_master_wr_mon_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

9.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.

- READY held LOW while gated is legal: a receiver may keep READY LOW until it chooses to accept (IHI 0022H §A3.2.1; IHI 0022L §A2.3.1). No paper (standard technique).
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

9.11 Navigation

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10 AXIL5 Slave Read

Module: `axil5_slave_rd.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_rd`

10.1 Overview

The AXIL5 Slave Read provides a buffered AXI5-Lite read interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_rd` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)

10.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate

Parameter	Type	Default	Description
			side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AR	int	2	
SKID_DEPTH_R	int	4	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
ARSize	int	AW + 3 +	
RSize	int	DW + 2 +	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

10.3 Ports

The s_axil_* side faces the bus; your backend drives the fub_* side.

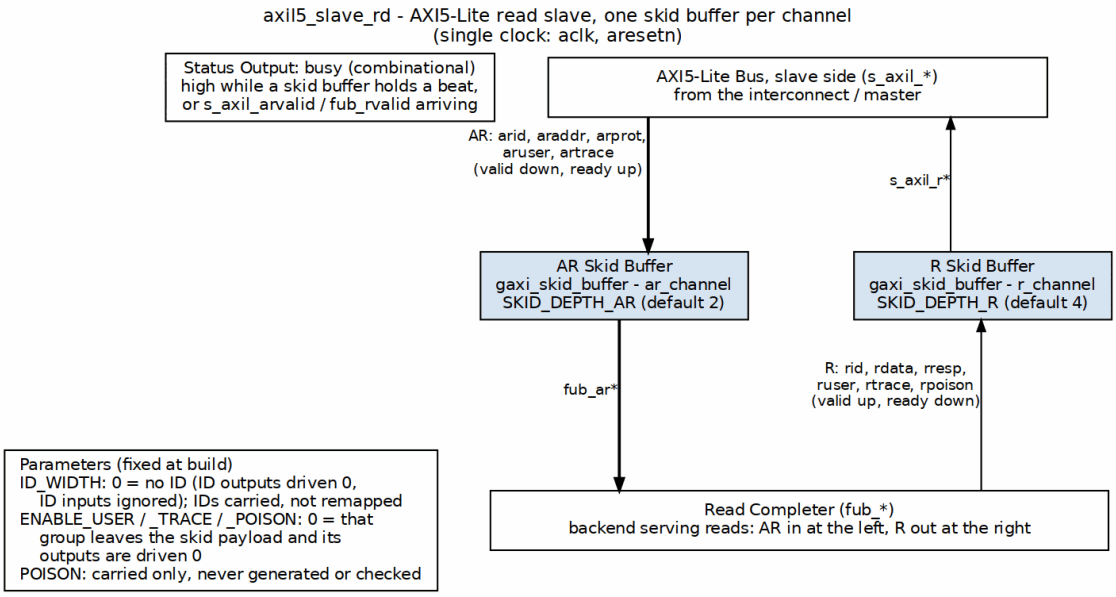
Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	

Port	Direction	Width	Description
s_axil_arid	Input	[IW-1:0]	
s_axil_araddr	Input	[AW-1:0]	
s_axil_arprot	Input	[2:0]	
s_axil_aruser	Input	[UW-1:0]	
s_axil_artrace	Input	1	
s_axil_arvalid	Input	1	
s_axil_arready	Output	1	
fub_arid	Output	[IW-1:0]	
fub_araddr	Output	[AW-1:0]	
fub_arprot	Output	[2:0]	
fub_aruser	Output	[UW-1:0]	
fub_artrace	Output	1	
fub_arvalid	Output	1	
fub_arready	Input	1	
fub_rid	Input	[IW-1:0]	
fub_rdata	Input	[DW-1:0]	
fub_rresp	Input	[1:0]	
fub_ruser	Input	[UW-1:0]	
fub_rtrace	Input	1	
fub_rpoison	Input	[PW-1:0]	
fub_rvalid	Input	1	
fub_rready	Output	1	
s_axil_rid	Output	[IW-1:0]	
s_axil_rdata	Output	[DW-1:0]	
s_axil_rresp	Output	[1:0]	
s_axil_ruser	Output	[UW-1:0]	
s_axil_rtrace	Output	1	
s_axil_rpoison	Output	[PW-1:0]	

Port	Direction	Width	Description
s_axil_rvalid	Output	1	
s_axil_rready	Input	1	
busy	Output	1	

10.4 Functional Description

10.4.1 Block diagram



axil5_slave_rd block diagram

The AXI5-Lite bus (`s_axil_*`) sits at the top and the completer (`fub_*`) is the bar at the bottom. AR runs down through its `gaxi_skid_buffer` to the bottom, and the R response climbs back up its own lane on the right through a second skid buffer, so every channel crosses exactly one buffer. This is the axil4 structure; AXI5-Lite only adds carried fields – the ID on every channel and the optional USER, TRACE and POISON groups, each dropped from the skid payload (outputs driven 0) when its `ENABLE_*` parameter is 0, as the Parameters legend states. `busy` is high while any skid buffer holds a beat or a valid is arriving (s_axil_arvalid, fub_rvalid), the same definition as every other AMBA base module.

10.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

10.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

10.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the

Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With `ID_WIDTH = 0` (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

10.5 Timing Characteristics

10.5.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AR`	2 entries	Skid depth on the AR channel
`SKID_DEPTH_R`	4 entries	Skid depth on the R channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

10.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

10.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_rd #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4)
) u_axil5_slave_rd (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // s_axil_ar
    .s_axil_arid         (s_axil_arid),
    // ... `s_axil_araddr`, `s_axil_arprot`, `s_axil_aruser`, +3 more
    // fub_ar
    .fub_arid            (fub_arid),
    // ... `fub_araddr`, `fub_arprot`, `fub_aruser`, +3 more
    // fub_r
    .fub_rid             (fub_rid),
    // ... `fub_rdata`, `fub_rresp`, `fub_ruser`, +4 more
    // s_axil_r
    .s_axil_rid          (s_axil_rid),
    // ... `s_axil_rdata`, `s_axil_rresp`, `s_axil_ruser`, +4 more
    // status
    .busy                (busy)
);
```

10.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS`` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/`DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

10.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_rd` – the AXI4-Lite counterpart
- AXI4-Lite modules

10.9 Testing

`val/amba/test_axil5_slave_rd.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

10.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. “Theory of Latency-Insensitive Design.” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 – relay stations: storage on a back-pressured channel adds latency without changing function.
- Chipressian. *Crack the Hardware Interview – from RTL Designers’ Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Bruno, F., Eschemann, G. *The FPGA Programming Handbook*, 2nd ed., Packt, 2024, ch. 8 “Introduction to AXI”, §8.3 “AXI4 interfaces (full and AXI-Lite)” – textbook treatment of the AXI-Lite channel set (secondary source).

- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

10.11 Navigation

AXI5-Lite index | AMBA overview

11 AXIL5 Slave Read, Clock-Gated

Module: `axil5_slave_rd_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_rd_cg`

11.1 Overview

The AXIL5 Slave Read, Clock-Gated provides a buffered AXI5-Lite read interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_rd_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- One `amba_clock_gate_ctrl` gating the whole inner module

11.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and ARID/RID width,

Parameter	Type	Default	Description
			1..32 (Subordinate side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AR	int	2	
SKID_DEPTH_R	int	4	
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

11.3 Ports

The s_axil_* side faces the bus; your backend drives the fub_* side. The cfg_cg_* inputs run the clock gate.

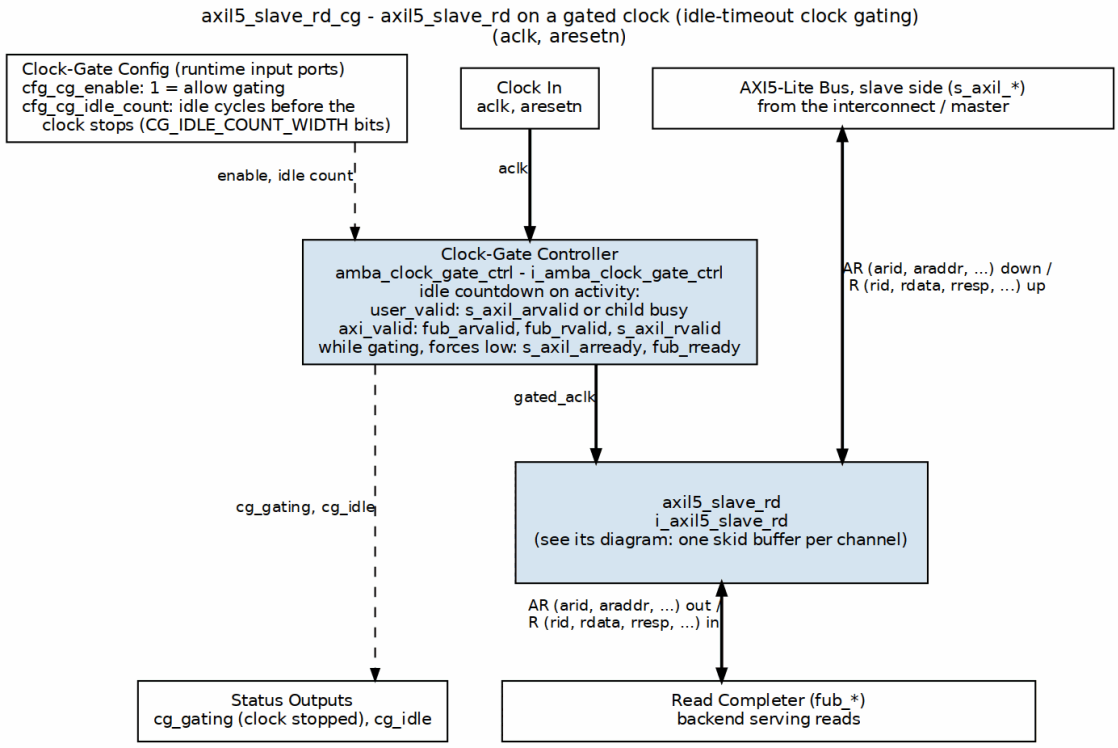
Port	Direction	Width	Description
aclk	Input	1	

Port	Direction	Width	Description
aresetn	Input	1	
cfg_cg_enable	Input	1	
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	
s_axil_arid	Input	[IW-1:0]	
s_axil_araddr	Input	[AW-1:0]	
s_axil_arprot	Input	[2:0]	
s_axil_aruser	Input	[UW-1:0]	
s_axil_artrace	Input	1	
s_axil_arvalid	Input	1	
s_axil_arready	Output	1	
fub_arid	Output	[IW-1:0]	
fub_araddr	Output	[AW-1:0]	
fub_arprot	Output	[2:0]	
fub_aruser	Output	[UW-1:0]	
fub_artrace	Output	1	
fub_arvalid	Output	1	
fub_arready	Input	1	
fub_rid	Input	[IW-1:0]	
fub_rdata	Input	[DW-1:0]	
fub_rresp	Input	[1:0]	
fub_ruser	Input	[UW-1:0]	
fub_rtrace	Input	1	
fub_rpoison	Input	[PW-1:0]	
fub_rvalid	Input	1	
fub_rready	Output	1	
s_axil_rid	Output	[IW-1:0]	
s_axil_rdata	Output	[DW-1:0]	
s_axil_rresp	Output	[1:0]	

Port	Direction	Width	Description
s_axil_ruser	Output	[UW-1:0]	
s_axil_rtrace	Output	1	
s_axil_rpoison	Output	[PW-1:0]	
s_axil_rvalid	Output	1	
s_axil_rready	Input	1	
cg_gating	Output	1	Active gating indicator
cg_idle	Output	1	All buffers empty indicator

11.4 Functional Description

11.4.1 Block diagram



axil5_slave_rd_cg block diagram

axil5_slave_rd is one child box (see its own diagram), clocked by gated_aclk. The heavy clock column runs from aclk through amba_clock_gate_ctrl, which counts down an idle window and stops the clock when nothing is active; its label lists the exact user_valid and axi_valid activity terms and the readies it forces low while gating, so no transfer is accepted on a stopped clock. cg_gating and cg_idle leave as status outputs.

11.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by ID_WIDTH); each group is gated by its own ENABLE_* parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

11.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

11.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

11.5 Timing Characteristics

11.5.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AR	2 entries	Skid depth on the AR channel
SKID_DEPTH_R	4 entries	Skid depth on the R channel

Each channel traverses one gaxi_skid_buffer. That module registers both rd_valid and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

11.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

11.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_rd_cg #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4),
    .CG_IDLE_COUNT_WIDTH (4)
) u_axil5_slave_rd_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // configuration
    .cfg_cg_enable       (cfg_cg_enable),
    .cfg_cg_idle_count   (cfg_cg_idle_count),
    // s_axil_ar
    .s_axil_arid         (s_axil_arid),
    // ... `s_axil_araddr`, `s_axil_arprot`, `s_axil_aruser`, +3 more
    // fub_ar
    .fub_arid            (fub_arid),
    // ... `fub_araddr`, `fub_arprot`, `fub_aruser`, +3 more
    // fub_r
    .fub_rid             (fub_rid),
    // ... `fub_rdata`, `fub_rresp`, `fub_ruser`, +4 more
    // s_axil_r
    .s_axil_rid          (s_axil_rid),
    // ... `s_axil_rdata`, `s_axil_rresp`, `s_axil_ruser`, +4 more
    // status
```

```
    .cg_gating          (cg_gating),
    .cg_idle            (cg_idle)
);
```

11.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS `` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/ `DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

A peer's READY must never enter the activity term. A consumer that parks its response-ready high while idle is behaving correctly; folding that signal into `user_valid` pins this block permanently awake and defeats gating entirely, silently, because function is unaffected. This wrapper wakes on VALIDs and pending work only. `val/amba/test_cg_peer_ready.py` parks the peer READY high, holds every VALID low, and requires `cg_gating` – see `vault/handbook/design/clock-gating-activity-terms.md`.

`busy` is not brought out here. The plain `_cg` wrappers consume the base module's `busy` internally as a wake term; use `cg_idle` or `cg_gating` for idle detection. The `_mon_cg` variants do expose `busy`.

`cfg_cg_enable` is not a kill switch. It arms gating and reaches `amba_clock_gate_ctrl` only. With it low the clock free-runs and the module behaves exactly like its base.

11.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_rd_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

11.9 Testing

`val/amba/test_axil5_slave_rd_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

11.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).

- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

11.11 Navigation

AXI5-Lite index | AMBA overview

12 AXIL5 Slave Read Monitor

Module: `axil5_slave_rd_mon.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_rd_mon`

12.1 Overview

The AXIL5 Slave Read Monitor provides a buffered AXI5-Lite read interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_rd_mon` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- Integrated transaction monitor emitting monbus packets

12.2 Parameters

A lite-monitor version of this wrapper exists as `axil5_slave_rd_monlite`: the same core and taps with `axi_monitor_lite` instead of `axi_monitor_filtered`, about a fifth of the monitor gates, error/timeout/completion/threshold packets only. Selected in generated bridges by `mon_preset = "lite"`.

Parameter	Type	Default	Description
`SKID_DEPTH_AR`	int	2	
`SKID_DEPTH_R`	int	4	
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and

Parameter	Type	Default	Description
			a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_MONITOR	bit	1'b1	0 = omit monitor, tie outputs
N_ADDR_RANGES	int	0	0 = address-range checker disabled
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
ID_FILTER_ENABLE	bit	1'b0	Per-instance ID-slice filter, inherited from the shared monitor core. Since amba #104 the wrapper feeds the AXI5-Lite ID, so the slice selects on AxID as on the AXI4 wrappers.
ADDR_FILTER_ENABLE	bit	1'b0	
ID_MATCH_BASE	int	0	First ID this

Parameter	Type	Default	Description
			instance owns when ID_FILTER_ENABLE =1. Compared against the AXI5-Lite AxID.
ID_MATCH_COUNT	int	0	Number of IDs owned from ID_MATCH_BASE. 0 = all IDs (the filter passes everything).
ACTIVE_TRANS_THRESHOLD	int	MAX_TRANSACTIONS / 2	
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.

Parameter	Type	Default	Description
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h0014	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h02	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit

Parameter	Type	Default	Description
			Unit ID for monitor packets)

The derived parameters (`AW`, `DW`, `UW`, … and the `*Size` payload widths) are computed from the ones above. **Do not override them.** Forcing a `*Size` to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how `test_axil5_master_wr` failed until `d6266344` removed those overrides.

12.3 Ports

The `s_axil_*` side faces the bus; your backend drives the `fub_axil_*` side. The `cfg_*` and `monbus_*` ports belong to the monitor.

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cam_clear`	Input	1	sync clear of the monitor trans CAM
`s_axil_arid`	Input	`[IW-1:0]`	
`s_axil_araddr`	Input	`[AW-1:0]`	
`s_axil_arprot`	Input	`[2:0]`	
`s_axil_aruser`	Input	`[UW-1:0]`	
`s_axil_artrace`	Input	1	
`s_axil_arvalid`	Input	1	
`s_axil_arready`	Output	1	
`s_axil_rid`	Output	`[IW-1:0]`	
`s_axil_rdata`	Output	`[DW-1:0]`	
`s_axil_rresp`	Output	`[1:0]`	
`s_axil_ruser`	Output	`[UW-1:0]`	
`s_axil_rtrace`	Output	1	
`s_axil_rpoison`	Output	`[PW-1:0]`	
`s_axil_rvalid`	Output	1	

Port	Direction	Width	Description
s_axil_rready	Input	1	
fub_axil_arid	Output	[IW-1:0]	
fub_axil_araddr	Output	[AW-1:0]	
fub_axil_arprot	Output	[2:0]	
fub_axil_aruser	Output	[UW-1:0]	
fub_axil_artrace	Output	1	
fub_axil_arvalid	Output	1	
fub_axil_arready	Input	1	
fub_axil_rid	Input	[IW-1:0]	
fub_axil_rdata	Input	[DW-1:0]	
fub_axil_rresp	Input	[1:0]	
fub_axil_ruser	Input	[UW-1:0]	
fub_axil_rtrace	Input	1	
fub_axil_rpoison	Input	[PW-1:0]	
fub_axil_rvalid	Input	1	
fub_axil_rready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets

Port	Direction	Width	Description
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_addr_check_enable	Input	1	

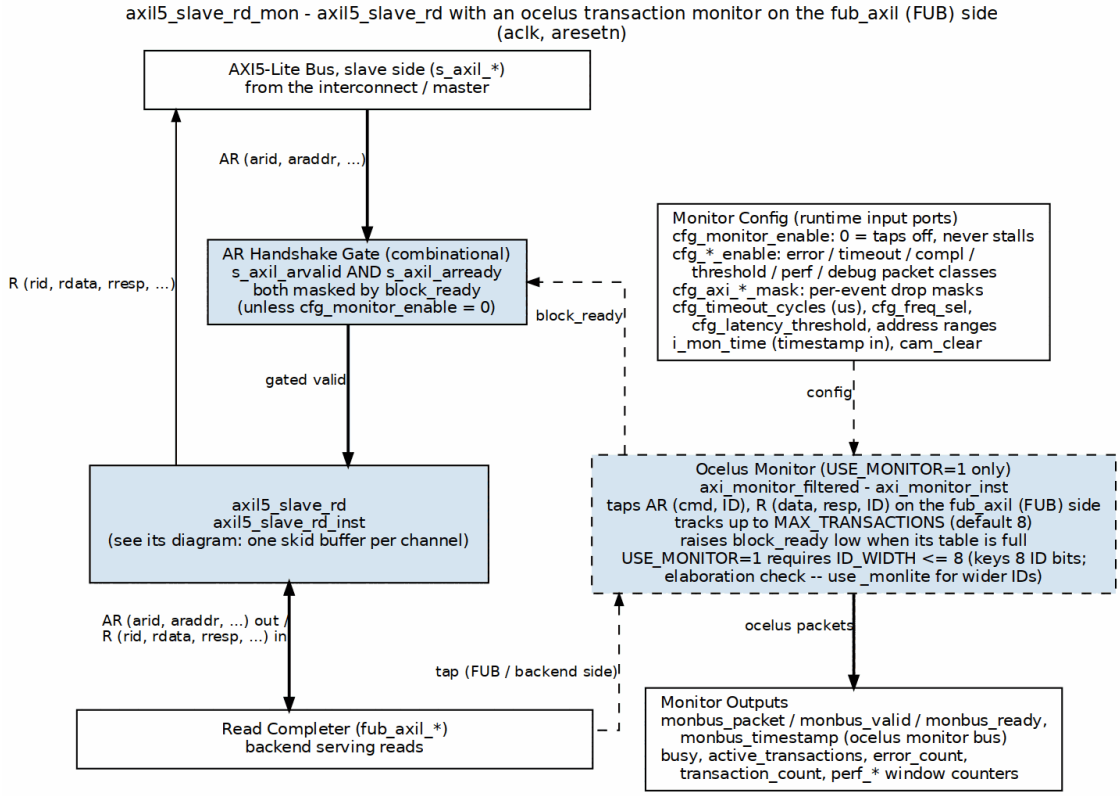
Port	Direction	Width	Description
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES >

Port	Direction	Width	Description
			0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
debug_block_ready	Output	1	
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	

Port	Direction	Width	Description
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

12.4 Functional Description

12.4.1 Block diagram



axil5_slave_rd_mon block diagram

`axil5_slave_rd` is one child box (see its own diagram). The AR request first passes a handshake gate that masks valid and ready with `block_ready`, so a full monitor table stalls new requests instead of losing them. The optional ocelus monitor (`axi_monitor_filtered`, dashed because `USE_MONITOR=0` removes it) taps the FUB-side (`fub_axil`) backend handshake below the core (every monitor taps the FUB side, for timing) while `block_ready` still gates the `s_axil` handshake, so the watchpoint trails the gate by the skid depth that the reserve margin covers; it carries the transaction IDs and emits monitor-bus packets as the heavy edge. With `USE_MONITOR=1` the build requires `ID_WIDTH <= 8`; the `_monlite` wrapper takes wider IDs.

12.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1
POISON	`ENABLE_POISON`	`WPOISON`, `RPOISON`	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

12.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, `ID_WIDTH=0`, groups off	AXI5-Lite defaults (`ID_WIDTH=4`, groups on)
`ARSize`	35	35	44
`RSize`	34	34	44
`AWSize`	35	35	44
`WSize`	36	36	41
`BSize`	2	2	11

(at `AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32` and the default widths. `ID_WIDTH = 0` is legal only on the Manager-side `axil5_master_*` modules; a Subordinate must carry the ID, so the `axil5_slave_*` modules require `ID_WIDTH >= 1` and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what `val/amba/test_axil5_axil4_equivalence.py` pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both `axil4_master_rd` and `axil5_master_rd` elaborated with `ID_WIDTH=0` and every `ENABLE_*` = 0.

12.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With `ID_WIDTH = 0` (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

12.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

12.6 Timing Characteristics

12.6.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AR	2 entries	Skid depth on the AR channel
SKID_DEPTH_R	4 entries	Skid depth on the R channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

12.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

12.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_rd_mon #(
    .SKID_DEPTH_AR      (2),
    .SKID_DEPTH_R       (4),
    .AXIL_ADDR_WIDTH    (32),
    .AXIL_DATA_WIDTH    (32),
    .ACLK_MHZ           (100),
    .USE_MONITOR        (1'b1),
    .N_ADDR_RANGES      (0),
    .UNIT_ID            (8'h02),
    .AGENT_ID           (16'h0014),
```

```
    .MAX_TRANSACTIONS    (8),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .ID_FILTER_ENABLE    (1'b0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ID_MATCH_BASE       (0),
    .ID_MATCH_COUNT      (0),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_slave_rd_mon (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_araddr`, `fub_axil_arprot`, `fub_axil_arlock`,
+31 more
    // s_axil_ar
    .s_axil_arid         (s_axil_arid),
    // ... `s_axil_araddr`, `s_axil_arprot`, `s_axil_aruser`, +3 more
    // s_axil_r
    .s_axil_rid          (s_axil_rid),
    // ... `s_axil_rdata`, `s_axil_rresp`, `s_axil_ruser`, +4 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +28 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

12.8 Related Modules

- axil5_master_rd

- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_rd_mon` – the AXI4-Lite counterpart
- AXI4-Lite modules

12.9 Testing

`val/amba/test_axil5_slave_rd_mon.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

12.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an

AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.

- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Chipressian. *Crack the Hardware Interview – from RTL Designers’ Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

12.11 Navigation

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13 AXIL5 Slave Read Monitor, Clock-Gated

Module: `axil5_slave_rd_mon_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_rd_mon_cg`

13.1 Overview

The AXIL5 Slave Read Monitor, Clock-Gated provides a buffered AXI5-Lite read interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_rd_mon_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite read path (AR and R channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- Integrated transaction monitor emitting monbus packets
- Monitor plus clock gating; ready signals held low while gated

13.2 Parameters

A lite-monitor version of this clock-gated wrapper exists as `axil5_slave_rd_monlite_cg`: the same gating logic around `axil5_slave_rd_monlite` instead of `axil5_slave_rd_mon`, about a fifth of the monitor gates.

Parameter	Type	Default	Description
`SKID_DEPTH_AR`	int	2	
`SKID_DEPTH_R`	int	4	
`AXIL_ADDR_WIDTH`	int	32	

Parameter	Type	Default	Description
AXIL_DATA_WIDTH	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
USE_MONITOR	bit	1'b1	0 = omit monitor in inner mon; outputs tied
N_ADDR_RANGES	int	0	0 = address-range checker disabled
ADDR_FILTER_ENABLE	bit	1'b0	0 = address filter inert
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOG	bit	1'b1	

Parameter	Type	Default	Description
IC			
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h0014	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to

Parameter	Type	Default	Description
			identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h02	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit Unit ID for monitor packets)

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

13.3 Ports

The s_axil_* side faces the bus; your backend drives the fub_axil_* side. The cfg_* and monbus_* ports belong to the monitor, and cfg_cg_* runs the clock gate.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
cam_clear	Input	1	sync clear of the monitor trans CAM
s_axil_arid	Input	[IW-1:0]	

Port	Direction	Width	Description
s_axil_araddr	Input	[AW-1:0]	
s_axil_arprot	Input	[2:0]	
s_axil_aruser	Input	[UW-1:0]	
s_axil_artrace	Input	1	
s_axil_arvalid	Input	1	
s_axil_arready	Output	1	
s_axil_rid	Output	[IW-1:0]	
s_axil_rdata	Output	[DW-1:0]	
s_axil_rresp	Output	[1:0]	
s_axil_ruser	Output	[UW-1:0]	
s_axil_rtrace	Output	1	
s_axil_rpoison	Output	[PW-1:0]	
s_axil_rvalid	Output	1	
s_axil_rready	Input	1	
fub_axil_arid	Output	[IW-1:0]	
fub_axil_araddr	Output	[AW-1:0]	
fub_axil_arprot	Output	[2:0]	
fub_axil_aruser	Output	[UW-1:0]	
fub_axil_artrace	Output	1	
fub_axil_arvalid	Output	1	
fub_axil_arready	Input	1	
fub_axil_rid	Input	[IW-1:0]	
fub_axil_rdata	Input	[DW-1:0]	
fub_axil_rresp	Input	[1:0]	
fub_axil_ruser	Input	[UW-1:0]	
fub_axil_rtrace	Input	1	
fub_axil_rpoison	Input	[PW-1:0]	
fub_axil_rvalid	Input	1	
fub_axil_rready	Output	1	
cfg_monitor_enable	Input	1	Enable

Port	Direction	Width	Description
			monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_ma	Input	[15:0]	Individual

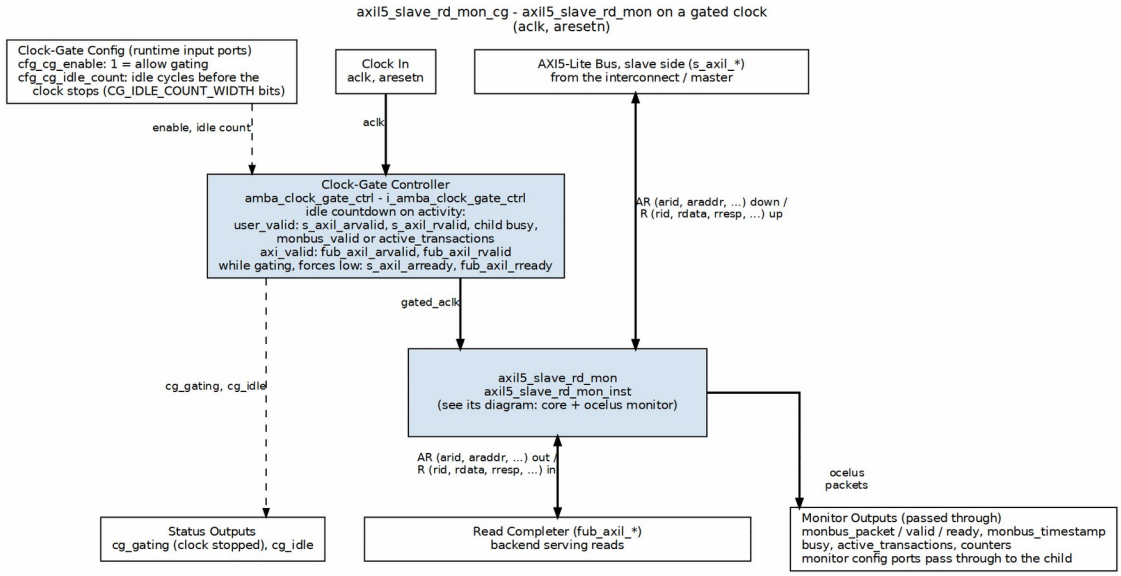
Port	Direction	Width	Description
sk			completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_cg_enable	Input	1	Enable clock gating
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	Idle cycles before gating
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when

Port	Direction	Width	Description
			monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
cg_gating	Output	1	Gated clock is

Port	Direction	Width	Description
			stopped
cg_idle	Output	1	No activity observed
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

13.4 Functional Description

13.4.1 Block diagram



axil5_slave_rd_mon_cg block diagram

`axil5_slave_rd_mon` (core plus ocelus monitor) is one child box (see its own diagram), clocked by `gated_aclk` from `amba_clock_gate_ctrl`. The controller's label lists the activity terms, which add `monbus_valid` and `active_transactions` so the clock keeps running while the monitor still has packets or open transactions. Monitor-bus packets and the monitor configuration pass straight through to the child.

13.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1

Group	Parameter	Signals on this module	Width
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

13.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

13.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them “not present” on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group’s output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

13.5 Design Notes

Four things worth knowing before you trust what comes out of the monitor:

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong, silently.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

13.6 Timing Characteristics

13.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AR`	2 entries	Skid depth on the AR channel
`SKID_DEPTH_R`	4 entries	Skid depth on the R channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the**

unstalled case – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

13.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

13.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_rd_mon_cg #(
    .SKID_DEPTH_AR       (2),
    .SKID_DEPTH_R        (4),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ACLK_MHZ            (100),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .UNIT_ID             (8'h02),
    .AGENT_ID            (16'h0014),
    .MAX_TRANSACTIONS    (8),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .CG_IDLE_COUNT_WIDTH (4),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
```

```
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_slave_rd_mon_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_araddr`, `fub_axil_arprot`, `fub_axil_arlock`,
+32 more
    // s_axil_ar
    .s_axil_arid         (s_axil_arid),
    // ... `s_axil_araddr`, `s_axil_arprot`, `s_axil_aruser`, +3 more
    // s_axil_r
    .s_axil_rid          (s_axil_rid),
    // ... `s_axil_rdata`, `s_axil_rresp`, `s_axil_ruser`, +4 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +30 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

13.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_rd_mon_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

13.9 Testing

`val/amba/test_axil5_slave_rd_mon_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in

`COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

13.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AR and R channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- READY held LOW while gated is legal: a receiver may keep READY LOW until it chooses to accept (IHI 0022H §A3.2.1; IHI 0022L §A2.3.1). No paper (standard technique).

- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

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14 AXIL5 Slave Write

Module: `axil5_slave_wr.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_wr`

14.1 Overview

The AXIL5 Slave Write provides a buffered AXI5-Lite write interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_wr` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)

14.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate

Parameter	Type	Default	Description
			side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AW	int	2	
SKID_DEPTH_W	int	4	
SKID_DEPTH_B	int	2	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
SW	int	DW / 8	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AWSize	int	AW + 3 +	
WSize	int	DW + SW +	
BSize	int	2 +	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

14.3 Ports

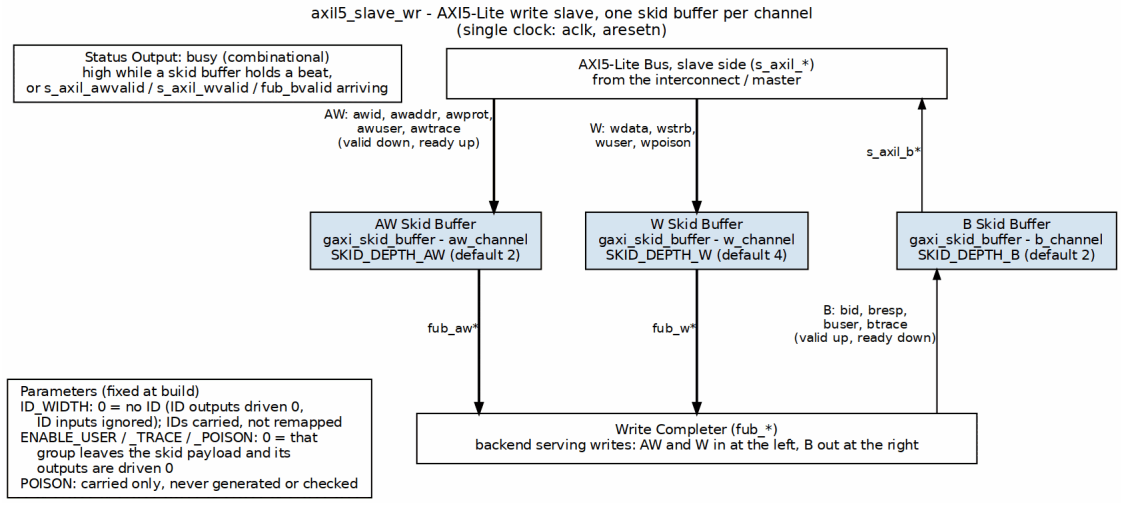
The s_axil_* side faces the bus; your backend drives the fub_* side.

Port	Direction	Width	Description
aclk	Input	1	
aresetn	Input	1	
s_axil_awid	Input	[IW-1:0]	
s_axil_awaddr	Input	[AW-1:0]	
s_axil_awprot	Input	[2:0]	
s_axil_awuser	Input	[UW-1:0]	
s_axil_awtrace	Input	1	
s_axil_awvalid	Input	1	
s_axil_awready	Output	1	
fub_awid	Output	[IW-1:0]	
fub_awaddr	Output	[AW-1:0]	
fub_awprot	Output	[2:0]	
fub_awuser	Output	[UW-1:0]	
fub_awtrace	Output	1	
fub_awvalid	Output	1	
fub_awready	Input	1	
s_axil_wdata	Input	[DW-1:0]	
s_axil_wstrb	Input	[SW-1:0]	
s_axil_wuser	Input	[UW-1:0]	
s_axil_wpoison	Input	[PW-1:0]	
s_axil_wvalid	Input	1	
s_axil_wready	Output	1	
fub_wdata	Output	[DW-1:0]	
fub_wstrb	Output	[SW-1:0]	
fub_wuser	Output	[UW-1:0]	
fub_wpoison	Output	[PW-1:0]	
fub_wvalid	Output	1	
fub_wready	Input	1	

Port	Direction	Width	Description
fub_bid	Input	[IW-1:0]	
fub_bresp	Input	[1:0]	
fub_buser	Input	[UW-1:0]	
fub_btrace	Input	1	
fub_bvalid	Input	1	
fub_bready	Output	1	
s_axil_bid	Output	[IW-1:0]	
s_axil_bresp	Output	[1:0]	
s_axil_buser	Output	[UW-1:0]	
s_axil_btrace	Output	1	
s_axil_bvalid	Output	1	
s_axil_bready	Input	1	
busy	Output	1	

14.4 Functional Description

14.4.1 Block diagram



axil5_slave_wr block diagram

The AXI5-Lite bus (`s_axil_*`) sits at the top and the completer (`fub_*`) is the bar at the bottom. AW and W run down through their own `gaxi_skid_buffers`, and the B response climbs back up its own lane on the right through a third, so every channel crosses exactly one buffer. This is the

axil4 structure; AXI5-Lite only adds carried fields – the ID on every channel and the optional USER, TRACE and POISON groups, each dropped from the skid payload (outputs driven 0) when its ENABLE_* parameter is 0, as the Parameters legend states. busy is high while any skid buffer holds a beat or a valid is arriving (s_axil_awvalid, s_axil_wvalid, fub_bvalid), the same definition as every other AMBA base module.

14.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by ID_WIDTH); each group is gated by its own ENABLE_* parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

14.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

14.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

14.5 Timing Characteristics

14.5.1 Buffer Depths and Latency

Parameter	Default	Channel
SKID_DEPTH_AW	2 entries	Skid depth on the AW channel
SKID_DEPTH_W	4 entries	Skid depth on the W channel
SKID_DEPTH_B	2 entries	Skid depth on the B channel

Each channel traverses one gaxi_skid_buffer. That module registers both rd_valid and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the

depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

14.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

14.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_wr #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (4),
    .SKID_DEPTH_B        (2)
) u_axil5_slave_wr (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // s_axil_aw
    .s_axil_awid         (s_axil_awid),
    // ... `s_axil_awaddr`, `s_axil_awprot`, `s_axil_awuser`, +3 more
    // fub_aw
    .fub_awid            (fub_awid),
    // ... `fub_awaddr`, `fub_awprot`, `fub_awuser`, +3 more
    // s_axil_w
    .s_axil_wdata        (s_axil_wdata),
    // ... `s_axil_wstrb`, `s_axil_wuser`, `s_axil_wpoison`, +2 more
    // fub_w
    .fub_wdata           (fub_wdata),
    // ... `fub_wstrb`, `fub_wuser`, `fub_wpoison`, +2 more
    // fub_b
```

```
    .fub_bid             (fub_bid),
    // ... `fub_bresp`, `fub_buser`, `fub_btrace`, +2 more
    // s_axil_b
    .s_axil_bid          (s_axil_bid),
    // ... `s_axil_bresp`, `s_axil_buser`, `s_axil_btrace`, +2 more
    // status
    .busy                (busy)
);
```

14.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS `` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/ `DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads “Extracting 4 bits from only 2 bit number”, and it is how this family’s test port broke.

The ID is AXI5-Lite’s, not the monitor’s. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

14.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon

- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_wr` – the AXI4-Lite counterpart
- AXI4-Lite modules

14.9 Testing

`val/amba/test_axil5_slave_wr.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

14.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*,

pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel's `gaxi_skid_buffer`.

- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. "Theory of Latency-Insensitive Design." *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 – relay stations: storage on a back-pressured channel adds latency without changing function.
- Chipressian. *Crack the Hardware Interview – from RTL Designers' Perspective: Architecture and Micro-architecture Design*, ch. 2 "Handshake Protocols", Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Bruno, F., Eschemann, G. *The FPGA Programming Handbook*, 2nd ed., Packt, 2024, ch. 8 "Introduction to AXI", §8.3 "AXI4 interfaces (full and AXI-Lite)" – textbook treatment of the AXI-Lite channel set (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

14.11 Navigation

AXI5-Lite index | AMBA overview

15 AXIL5 Slave Write, Clock-Gated

Module: `axil5_slave_wr_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_wr_cg`

15.1 Overview

The AXIL5 Slave Write, Clock-Gated provides a buffered AXI5-Lite write interface for slave devices.

AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_wr_cg` with the ID and those groups threaded through the packed SKID payload.

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- One `amba_clock_gate_ctrl` gating the whole inner module

15.2 Parameters

Parameter	Type	Default	Description
`AXIL_ADDR_WIDTH`	int	32	
`AXIL_DATA_WIDTH`	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
`ID_WIDTH`	int	4	AWID/BID and

Parameter	Type	Default	Description
			ARID/RID width, 1..32 (Subordinate side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
SKID_DEPTH_AW	int	2	
SKID_DEPTH_W	int	4	
SKID_DEPTH_B	int	2	
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
SW	int	DW / 8	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	

The derived parameters (AW, DW, UW, … and the *Size payload widths) are computed from the ones above. **Do not override them.** Forcing a *Size to a value the RTL did not derive makes every optional-group field a part-select past the end of the vector – which is exactly how test_axil5_master_wr failed until d6266344 removed those overrides.

15.3 Ports

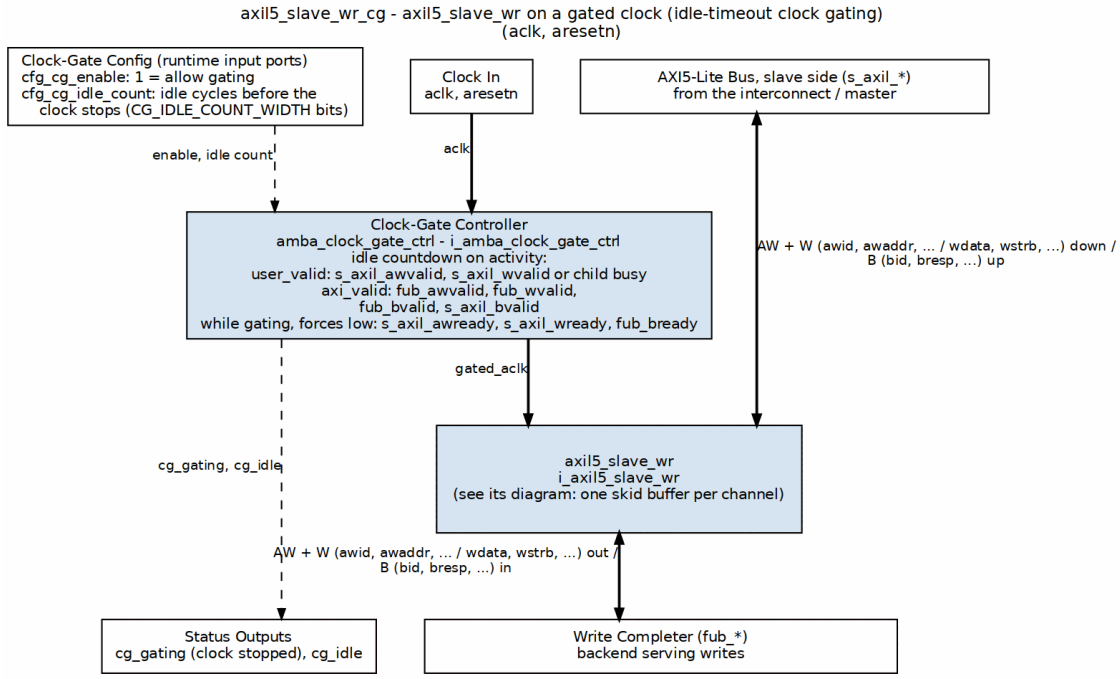
The `s_axil_*` side faces the bus; your backend drives the `fub_*` side. The `cfg_cg_*` inputs run the clock gate.

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cfg_cg_enable`	Input	1	
`cfg_cg_idle_count`	Input	`[CG_IDLE_COUNT_WIDTH-1:0]`	
`s_axil_awid`	Input	`[IW-1:0]`	
`s_axil_awaddr`	Input	`[AW-1:0]`	
`s_axil_awprot`	Input	`[2:0]`	
`s_axil_awuser`	Input	`[UW-1:0]`	
`s_axil_awtrace`	Input	1	
`s_axil_awvalid`	Input	1	
`s_axil_awready`	Output	1	
`fub_awid`	Output	`[IW-1:0]`	
`fub_awaddr`	Output	`[AW-1:0]`	
`fub_awprot`	Output	`[2:0]`	
`fub_awuser`	Output	`[UW-1:0]`	
`fub_awtrace`	Output	1	
`fub_awvalid`	Output	1	
`fub_awready`	Input	1	
`s_axil_wdata`	Input	`[DW-1:0]`	
`s_axil_wstrb`	Input	`[SW-1:0]`	
`s_axil_wuser`	Input	`[UW-1:0]`	
`s_axil_wpoison`	Input	`[PW-1:0]`	
`s_axil_wvalid`	Input	1	
`s_axil_wready`	Output	1	

Port	Direction	Width	Description
fub_wdata	Output	[DW-1:0]	
fub_wstrb	Output	[SW-1:0]	
fub_wuser	Output	[UW-1:0]	
fub_wpoison	Output	[PW-1:0]	
fub_wvalid	Output	1	
fub_wready	Input	1	
fub_bid	Input	[IW-1:0]	
fub_bresp	Input	[1:0]	
fub_buser	Input	[UW-1:0]	
fub_btrace	Input	1	
fub_bvalid	Input	1	
fub_bready	Output	1	
s_axil_bid	Output	[IW-1:0]	
s_axil_bresp	Output	[1:0]	
s_axil_buser	Output	[UW-1:0]	
s_axil_btrace	Output	1	
s_axil_bvalid	Output	1	
s_axil_bready	Input	1	
cg_gating	Output	1	Active gating indicator
cg_idle	Output	1	All buffers empty indicator

15.4 Functional Description

15.4.1 Block diagram



axil5_slave_wr_cg block diagram

`axil5_slave_wr` is one child box (see its own diagram), clocked by `gated_aclk`. The heavy clock column runs from `aclk` through `amba_clock_gate_ctrl`, which counts down an idle window and stops the clock when nothing is active; its label lists the exact `user_valid` and `axi_valid` activity terms and the readies it forces low while gating, so no transfer is accepted on a stopped clock. `cg_gating` and `cg_idle` leave as status outputs.

15.4.2 Optional signal groups

The ID plus three optional groups – the ones IHI 0022L Table B2.2 allows on AXI5-Lite. The ID has no enable (it is sized by `ID_WIDTH`); each group is gated by its own `ENABLE_*` parameter and contributes to the packed SKID payload only when enabled.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`

Group	Parameter	Signals on this module	Width
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

15.4.3 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

15.4.4 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them “not present” on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group’s output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

15.5 Timing Characteristics

15.5.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	4 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

15.5.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

15.6 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_wr_cg #(
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1),
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (4),
    .SKID_DEPTH_B        (2),
    .CG_IDLE_COUNT_WIDTH (4)
) u_axil5_slave_wr_cg (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // configuration
    .cfg_cg_enable       (cfg_cg_enable),
    .cfg_cg_idle_count   (cfg_cg_idle_count),
    // s_axil_aw
    .s_axil_awid         (s_axil_awid),
    // ... `s_axil_awaddr`, `s_axil_awprot`, `s_axil_awuser`, +3 more
    // fub_aw
    .fub_awid            (fub_awid),
    // ... `fub_awaddr`, `fub_awprot`, `fub_awuser`, +3 more
    // s_axil_w
    .s_axil_wdata        (s_axil_wdata),
    // ... `s_axil_wstrb`, `s_axil_wuser`, `s_axil_wpoison`, +2 more
    // fub_w
    .fub_wdata           (fub_wdata),
    // ... `fub_wstrb`, `fub_wuser`, `fub_wpoison`, +2 more
    // fub_b
    .fub_bid             (fub_bid),
    // ... `fub_bresp`, `fub_buser`, `fub_btrace`, +2 more
    // s_axil_b
    .s_axil_bid          (s_axil_bid),
    // ... `s_axil_bresp`, `s_axil_buser`, `s_axil_btrace`, +2 more
    // status
    .cg_gating           (cg_gating),
    .cg_idle             (cg_idle)
);
```

15.7 Design Notes

No ID and all optional groups off is AXI4-Lite. `ARSize`/`AWSize`/`WSize`/`RSize`/ `BSize` are sums of `ID_WIDTH` and conditional terms over the `ENABLE_*` parameters. With `ID_WIDTH=0` (Manager

side only) and every group disabled they collapse to the AXI4-Lite payload widths exactly, which is what `val/amba/test_axil5_axil4_equivalence.py` asserts against both the AXI4-Lite RTL and `axil5_master_rd` configured that way.

Illegal widths do not elaborate. An `initial begin : param_check` block under `` `ifndef SYNTHESIS`` stops the build when `AXIL_DATA_WIDTH` is not an AXI data width (8..1024, power of two; IHI 0022L Table A3.19 – Table B2.4 adds no AXI5-Lite restriction) or `ID_WIDTH` is outside 0..32 (1..32 on the slave-side modules). `val/amba/test_axil_param_guard.py` checks it.

Do not override a derived parameter. The `*Size` parameters and the `AW`/`DW`/`IW` aliases are declared as `parameter` so the elaborator can compute them, not so callers can set them. Forcing one to the AXI4-Lite value while a group is enabled part-selects past the end of the vector – the failure reads "Extracting 4 bits from only 2 bit number", and it is how this family's test port broke.

The ID is AXI5-Lite's, not the monitor's. AXI5-Lite carries AWID/BID and ARID/RID (IHI 0022L Table B2.2, code YS: optional on a Manager, mandatory on a Subordinate) and this family carries them through. The `_mon`/`_monlite` wrappers feed it to the shared monitor core (amba #104), so each response is paired with the oldest outstanding command of its own ID and a Subordinate may reorder across IDs (B2.1.5); W carries no ID and is attributed in AW order (A5.5).

The skid buffer is not a bypass. `gaxi_skid_buffer` registers both `rd_valid` and its storage, so every transfer costs one cycle even unstalled. Depth buys backpressure absorption, not throughput.

A peer's READY must never enter the activity term. A consumer that parks its response-ready high while idle is behaving correctly; folding that signal into `user_valid` pins this block permanently awake and defeats gating entirely, silently, because function is unaffected. This wrapper wakes on VALIDs and pending work only. `val/amba/test_cg_peer_ready.py` parks the peer READY high, holds every VALID low, and requires `cg_gating` – see `vault/handbook/design/clock-gating-activity-terms.md`.

`busy` is not brought out here. The plain `_cg` wrappers consume the base module's `busy` internally as a wake term; use `cg_idle` or `cg_gating` for idle detection. The `_mon_cg` variants do expose `busy`.

`cfg_cg_enable` is not a kill switch. It arms gating and reaches `amba_clock_gate_ctrl` only. With it low the clock free-runs and the module behaves exactly like its base.

15.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr

- axil5_master_wr_cg
- `axil4_slave_wr_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

15.9 Testing

`val/amba/test_axil5_slave_wr_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** – `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically.

15.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 “AXI5-Lite” and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. “A Survey of Design Techniques for System-Level Dynamic Power Management.” *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.

- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

15.11 Navigation

AXI5-Lite index | AMBA overview

16 AXIL5 Slave Write Monitor

Module: `axil5_slave_wr_mon.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_wr_mon`

16.1 Overview

The AXIL5 Slave Write Monitor gives a slave device a buffered AXI5-Lite write interface — and keeps a transaction monitor on the path while it's at it.

If you take one thing from this page, take this: AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_wr_mon` with the ID and those groups threaded through the packed SKID payload. Know the AXI4-Lite part and you know this one.

16.1.1 Key Features

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- Integrated transaction monitor emitting monbus packets

16.2 Parameters

A lite-monitor version of this wrapper exists as `axil5_slave_wr_monlite`: the same core and taps with `axi_monitor_lite` instead of `axi_monitor_filtered`, about a fifth of the monitor gates, error/timeout/completion/threshold packets only. Selected in generated bridges by `mon_preset = "lite"`.

Parameter	Type	Default	Description
`SKID_DEPTH_AW`	int	2	
`SKID_DEPTH_W`	int	2	

Parameter	Type	Default	Description
SKID_DEPTH_B	int	2	
AXIL_ADDR_WIDTH	int	32	
AXIL_DATA_WIDTH	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_MONITOR	bit	1'b1	0 = omit monitor, tie outputs
N_ADDR_RANGES	int	0	0 = address-range checker disabled
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
ID_FILTER_ENABLE	bit	1'b0	Per-instance ID-slice filter, inherited from the shared monitor core. Since amba #104 the wrapper feeds the AXI5-Lite ID, so the slice selects on AxID as on the

Parameter	Type	Default	Description
			AXI4 wrappers.
ADDR_FILTER_ENABLE	bit	1'b0	
ID_MATCH_BASE	int	0	First ID this instance owns when ID_FILTER_ENABLE=1. Compared against the AXI5-Lite AxID.
ID_MATCH_COUNT	int	0	Number of IDs owned from ID_MATCH_BASE. 0 = all IDs (the filter passes everything).
ACTIVE_TRANS_THRESHOLD	int	MAX_TRANSACTIONS / 2	
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate

Parameter	Type	Default	Description
			side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h0015	Agent identifier emitted in the agent_id field of every monitor packet. Pairs with UNIT_ID to identify the packet source. (16-bit Agent ID for monitor packets)
UNIT_ID		8'h02	Unit identifier emitted in the unit_id field of every monitor packet. Give each monitored interface a distinct value or

Parameter	Type	Default	Description
			the packets cannot be told apart at the collector. (8-bit Unit ID for monitor packets)

The derived parameters (`AW`, `DW`, `UW`, … and the `*Size` payload widths) are computed from the ones above. **Do not override them.** Force a `*Size` to a value the RTL didn't derive and every optional-group field turns into a part-select past the end of the vector — which is exactly how `test_axil5_master_wr` failed until `d6266344` removed those overrides.

16.3 Ports

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cam_clear`	Input	1	sync clear of the monitor trans CAM
`s_axil_awid`	Input	`[IW-1:0]`	
`s_axil_awaddr`	Input	`[AW-1:0]`	
`s_axil_awprot`	Input	`[2:0]`	
`s_axil_awuser`	Input	`[UW-1:0]`	
`s_axil_awtrace`	Input	1	
`s_axil_awvalid`	Input	1	
`s_axil_awready`	Output	1	
`s_axil_wdata`	Input	`[DW-1:0]`	
`s_axil_wstrb`	Input	`[DW/8-1:0]`	
`s_axil_wuser`	Input	`[UW-1:0]`	
`s_axil_wpoison`	Input	`[PW-1:0]`	
`s_axil_wvalid`	Input	1	
`s_axil_wready`	Output	1	

Port	Direction	Width	Description
s_axil_bid	Output	[IW-1:0]	
s_axil_bresp	Output	[1:0]	
s_axil_buser	Output	[UW-1:0]	
s_axil_btrace	Output	1	
s_axil_bvalid	Output	1	
s_axil_bready	Input	1	
fub_axil_awid	Output	[IW-1:0]	
fub_axil_awaddr	Output	[AW-1:0]	
fub_axil_awprot	Output	[2:0]	
fub_axil_awuser	Output	[UW-1:0]	
fub_axil_awtrace	Output	1	
fub_axil_awvalid	Output	1	
fub_axil_awready	Input	1	
fub_axil_wdata	Output	[DW-1:0]	
fub_axil_wstrb	Output	[DW/8-1:0]	
fub_axil_wuser	Output	[UW-1:0]	
fub_axil_wpoison	Output	[PW-1:0]	
fub_axil_wvalid	Output	1	
fub_axil_wready	Input	1	
fub_axil_bid	Input	[IW-1:0]	
fub_axil_bresp	Input	[1:0]	
fub_axil_buser	Input	[UW-1:0]	
fub_axil_btrace	Input	1	
fub_axil_bvalid	Input	1	
fub_axil_bready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection

Port	Direction	Width	Description
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask

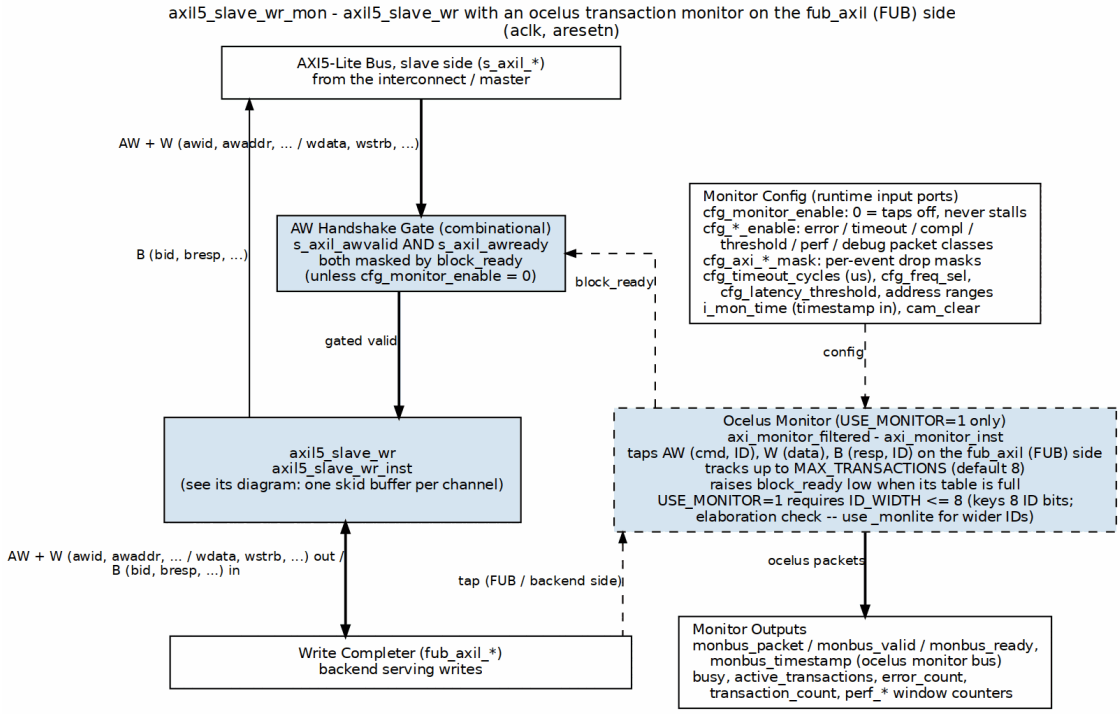
Port	Direction	Width	Description
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the

Port	Direction	Width	Description
			group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_low.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
debug_block_ready	Output	1	
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force	Input	1	

Port	Direction	Width	Description
_close			
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

16.4 Functional Description

16.4.1 Block diagram



axil5_slave_wr_mon block diagram

axil5_slave_wr is one child box (see its own diagram). The AW request first passes a handshake gate that masks valid and ready with block_ready, so a full monitor table stalls new requests instead of losing them. The optional ocelus monitor (axi_monitor_filtered, dashed because USE_MONITOR=0 removes it) taps the FUB-side (fub_axil) backend handshake below the core (every monitor taps the FUB side, for timing) while block_ready still gates the s_axil handshake, so the watchpoint trails the gate by the skid depth that the reserve margin covers; it carries the transaction IDs and emits monitor-bus packets as the heavy edge. With USE_MONITOR=1 the build requires ID_WIDTH <= 8; the _monlite wrapper takes wider IDs.

AXI5-Lite adds the transaction ID (ID_WIDTH, no enable; mandatory on this Subordinate-side module) and three optional signal groups – USER, TRACE and POISON, the ones IHI 0022L Table B2.2 allows – each gated by its own ENABLE_* parameter. A group contributes to the packed SKID payload only when it's enabled — turn it off and those bits simply aren't in the vector.

Group	Parameter	Signals on this module	Width
ID	ID_WIDTH (no enable)	AxID, BID, RID	ID_WIDTH (0..32; >= 1 on slave-side modules)
USER	ENABLE_USER	AxUSER, WUSER, BUSER, RUSER	USER_WIDTH
TRACE	ENABLE_TRACE	AxTRACE, BTRACE, RTRACE	1
POISON	ENABLE_POISON	WPOISON, RPOISON	one bit per 64 data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

16.4.2 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, ID_WIDTH=0, groups off	AXI5-Lite defaults (ID_WIDTH=4, groups on)
ARSize	35	35	44
RSize	34	34	44
AWSize	35	35	44
WSize	36	36	41
BSize	2	2	11

(at AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32 and the default widths. ID_WIDTH = 0 is legal only on the Manager-side axil5_master_* modules; a Subordinate must carry the ID, so the axil5_slave_* modules require ID_WIDTH >= 1 and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what val/amba/test_axil5_axil4_equivalence.py pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both axil4_master_rd and axil5_master_rd elaborated with ID_WIDTH=0 and every ENABLE_* = 0.

16.4.3 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With ID_WIDTH = 0 (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind ENABLE_* parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

16.5 Design Notes

A handful of traps specific to the monitored variants. Each one is written down because someone hit it.

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to axi_monitor_filtered, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so ID_WIDTH > 8 is refused at elaboration here (the _monlite wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

ACLK_MHZ is not decoration. It builds the microsecond tick LUT in counter_freq_invariant. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong — silently. That's the worst kind of wrong.

NUM_BANKS > 1 on a WRITE monitor requires USE_WDATA_ORDER_Q = 1.
`axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination. At least that one fails loud.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

16.6 Timing Characteristics

16.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	2 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

16.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

16.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_wr_mon #(
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (2),
    .SKID_DEPTH_B        (2),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .ACLK_MHZ            (100),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .UNIT_ID             (8'h02),
    .AGENT_ID            (16'h0015),
    .MAX_TRANSACTIONS    (8),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .ID_FILTER_ENABLE    (1'b0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ID_MATCH_BASE       (0),
    .ID_MATCH_COUNT      (0),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .ID_WIDTH            (4),
    .USER_WIDTH          (4),
    .ENABLE_USER         (1),
    .ENABLE_TRACE        (1),
    .ENABLE_POISON       (1)
) u_axil5_slave_wr_mon (
    // clock/reset
    .aclk                (aclk),
    .aresetn             (aresetn),
    // status
    .cam_clear           (cam_clear),
    // ... `fub_axil_awaddr`, `fub_axil_awprot`, `fub_axil_awlock`,
+35 more
    // s_axil_aw
    .s_axil_awid         (s_axil_awid),
    // ... `s_axil_awaddr`, `s_axil_awprot`, `s_axil_awuser`, +3 more
    // s_axil_w
    .s_axil_wdata        (s_axil_wdata),
    // ... `s_axil_wstrb`, `s_axil_wuser`, `s_axil_wpoison`, +2 more
    // s_axil_b
    .s_axil_bid          (s_axil_bid),
    // ... `s_axil_bresp`, `s_axil_buser`, `s_axil_btrace`, +2 more
```

```
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +28 more
    // monitor bus
    .monbus_valid        (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

16.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_wr_mon` – the AXI4-Lite counterpart
- AXI4-Lite modules

16.9 Testing

`val/amba/test_axil5_slave_wr_mon.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** — `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake — and the version you'd rather get.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically. That's exactly how you want a variant family to work.

16.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 "AXI5-Lite" and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling,

§A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).

- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 “AMBA AXI5-Lite”, §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Cortadella, J., Kishinevsky, M., Grundmann, B. “Synthesis of Synchronous Elastic Architectures.” *Proc. 43rd Design Automation Conf. (DAC ’06)*, pp. 657-662, 2006. DOI: 10.1145/1146909.1147077 – the elastic buffer; its Property 3.1 (capacity C >= 2 for full throughput) is the depth-2 floor of each channel’s `gaxi_skid_buffer`.
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Chipressian. *Crack the Hardware Interview – from RTL Designers’ Perspective: Architecture and Micro-architecture Design*, ch. 2 “Handshake Protocols”, Q8 valid-ready protocol and Q9 valid-ready slices – textbook treatment of register slices (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `ElasticBuffer__Cortadella2006.pdf`)

16.11 Navigation

AXI5-Lite index | AMBA overview

17 AXIL5 Slave Write Monitor, Clock-Gated

Module: `axil5_slave_wr_mon_cg.sv` **Location:** `rtl/amba/axil5/` **AXI4-Lite counterpart:** `axil4_slave_wr_mon_cg`

17.1 Overview

The AXIL5 Slave Write Monitor, Clock-Gated gives a slave device a buffered AXI5-Lite write interface — the same monitor as the plain variant, with clock gating stacked on top and the ready signals held low while gated.

If you take one thing from this page, take this: AXI5-Lite is AXI4-Lite plus the transaction ID and a few optional signal groups. It changes no channel's handshake, ordering or response semantics, so this module is structurally `axil4_slave_wr_mon_cg` with the ID and those groups threaded through the packed SKID payload. Know the AXI4-Lite part and you know this one.

17.1.1 Key Features

- AXI5-Lite write path (AW, W and B channels)
- Single-beat transactions: AXI-Lite has no burst support
- Configurable skid buffers on every channel for timing closure
- Elaboration fails on an illegal `AXIL_DATA_WIDTH` or `ID_WIDTH`
- Transaction ID (`AWID`/`BID`, `ARID`/`RID`; `ID_WIDTH`) plus three optional signal groups (USER, TRACE, POISON), each independently enabled
- With all groups disabled, the AXI4-Lite module of the same name plus the ID (mandatory on a Subordinate)
- Integrated transaction monitor emitting monbus packets
- Monitor plus clock gating; ready signals held low while gated

17.2 Parameters

A lite-monitor version of this clock-gated wrapper exists as `axil5_slave_wr_monlite_cg`: the same gating logic around `axil5_slave_wr_monlite` instead of `axil5_slave_wr_mon`, about a fifth of the monitor gates.

Parameter	Type	Default	Description
`SKID_DEPTH_AW`	int	2	
`SKID_DEPTH_W`	int	2	

Parameter	Type	Default	Description
SKID_DEPTH_B	int	2	
AXIL_ADDR_WIDTH	int	32	
AXIL_DATA_WIDTH	int	32	Any AXI data width, 8..1024 and a power of two (IHI 0022L Table A3.19; Table B2.4 adds no AXI5-Lite limit). Checked at elaboration.
USE_MONITOR	bit	1'b1	0 = omit monitor in inner mon; outputs tied
N_ADDR_RANGES	int	0	0 = address-range checker disabled
ADDR_FILTER_ENABLE	bit	1'b0	0 = address filter inert
ACLK_MHZ	int	100	
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	
USE_WDATA_ORDER_Q	bit	1'b0	
NUM_BANKS	int	1	
MAX_TRANSACTIONS	int	8	Maximum outstanding transactions (reduced for AXIL)
ENABLE_FILTERING	bit	1	Enable packet filtering
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
ENABLE_ERROR_LOGIC	bit	1'b1	

Parameter	Type	Default	Description
ENABLE_TIMEOUT_LOGIC	bit	1'b1	
ENABLE_COMPL_LOGIC	bit	1'b1	
ENABLE_THRESHOLD_LOGIC	bit	1'b1	
ENABLE_PERF_LOGIC	bit	1'b1	
ENABLE_DEBUG_LOGIC	bit	1'b0	
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown
ID_WIDTH	int	4	AWID/BID and ARID/RID width, 1..32 (Subordinate side: the ID is mandatory, Table B2.2 code YS). IHI 0022L Table A5.2.
USER_WIDTH	int	4	
ENABLE_USER	bit	1	
ENABLE_TRACE	bit	1	
ENABLE_POISON	bit	1	
AW	int	AXIL_ADDR_WIDTH	
DW	int	AXIL_DATA_WIDTH	
IW	int	(ID_WIDTH > 0) ? ID_WIDTH : 1	ID port width; a 1-bit placeholder when ID_WIDTH = 0
UW	int	USER_WIDTH	
PW	int	(DW / 64) > 0 ? (DW / 64) : 1	
AGENT_ID		16'h0015	Agent identifier emitted in the agent_id field of every monitor

Parameter	Type	Default	Description
			packet. Pairs with `UNIT_ID` to identify the packet source. (16-bit Agent ID for monitor packets)
`UNIT_ID`		`8'h02`	Unit identifier emitted in the `unit_id` field of every monitor packet. Give each monitored interface a distinct value or the packets cannot be told apart at the collector. (8-bit Unit ID for monitor packets)

The derived parameters (`AW`, `DW`, `UW`, … and the `*Size` payload widths) are computed from the ones above. **Do not override them.** Force a `*Size` to a value the RTL didn't derive and every optional-group field turns into a part-select past the end of the vector — which is exactly how `test_axil5_master_wr` failed until `d6266344` removed those overrides.

17.3 Ports

Port	Direction	Width	Description
`aclk`	Input	1	
`aresetn`	Input	1	
`cam_clear`	Input	1	sync clear of the monitor trans CAM
`s_axil_awid`	Input	`[IW-1:0]`	

Port	Direction	Width	Description
s_axil_awaddr	Input	[AW-1:0]	
s_axil_awprot	Input	[2:0]	
s_axil_awuser	Input	[UW-1:0]	
s_axil_awtrace	Input	1	
s_axil_awvalid	Input	1	
s_axil_awready	Output	1	
s_axil_wdata	Input	[DW-1:0]	
s_axil_wstrb	Input	[DW/8-1:0]	
s_axil_wuser	Input	[UW-1:0]	
s_axil_wpoison	Input	[PW-1:0]	
s_axil_wvalid	Input	1	
s_axil_wready	Output	1	
s_axil_bid	Output	[IW-1:0]	
s_axil_bresp	Output	[1:0]	
s_axil_buser	Output	[UW-1:0]	
s_axil_btrace	Output	1	
s_axil_bvalid	Output	1	
s_axil_bready	Input	1	
fub_axil_awid	Output	[IW-1:0]	
fub_axil_awaddr	Output	[AW-1:0]	
fub_axil_awprot	Output	[2:0]	
fub_axil_awuser	Output	[UW-1:0]	
fub_axil_awtrace	Output	1	
fub_axil_awvalid	Output	1	
fub_axil_awready	Input	1	
fub_axil_wdata	Output	[DW-1:0]	
fub_axil_wstrb	Output	[DW/8-1:0]	
fub_axil_wuser	Output	[UW-1:0]	
fub_axil_wpoison	Output	[PW-1:0]	
fub_axil_wvalid	Output	1	

Port	Direction	Width	Description
fub_axil_wready	Input	1	
fub_axil_bid	Input	[IW-1:0]	
fub_axil_bresp	Input	[1:0]	
fub_axil_buser	Input	[UW-1:0]	
fub_axil_btrace	Input	1	
fub_axil_bvalid	Input	1	
fub_axil_bready	Output	1	
cfg_monitor_enable	Input	1	Enable monitoring
cfg_error_enable	Input	1	Enable error detection
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_perf_enable	Input	1	Enable performance monitoring
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_debug_enable	Input	1	Enable debug packets
cfg_timeout_cycles	Input	[15:0]	Timeout threshold in MICROSECONDS (1 us tick), despite the name
cfg_freq_sel	Input	[3:0]	counter_freq_invariant LUT index
cfg_latency_threshold	Input	[31:0]	Latency threshold for alerts
cfg_axi_pkt_mask	Input	[15:0]	Drop mask for

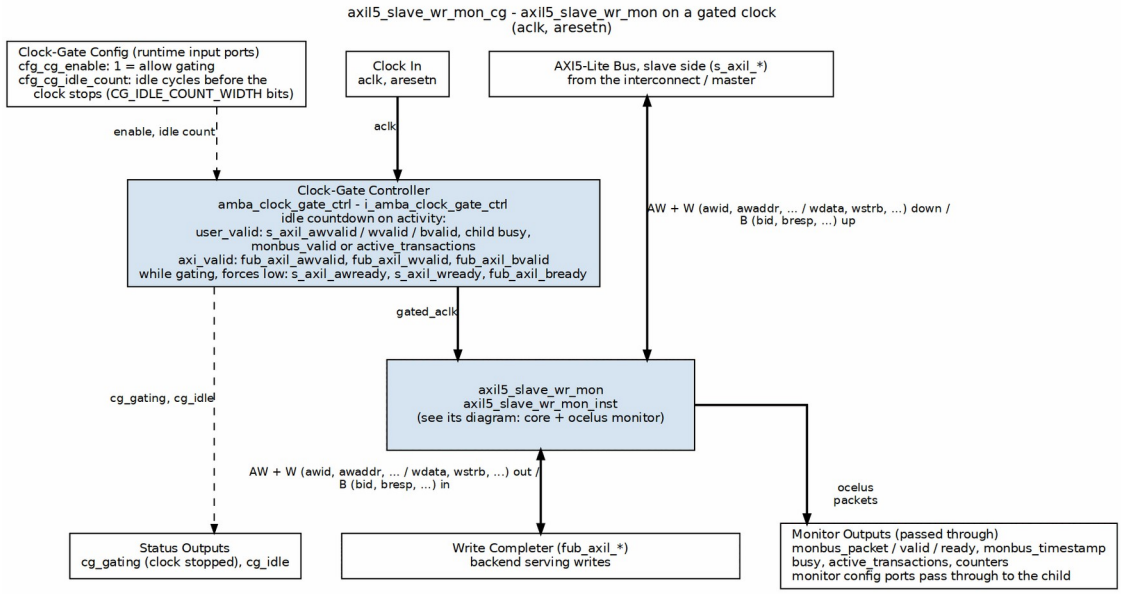
Port	Direction	Width	Description
			packet types
cfg_axi_err_select	Input	[15:0]	Error select for packet types
cfg_axi_error_mask	Input	[15:0]	Individual error event mask
cfg_axi_timeout_mask	Input	[15:0]	Individual timeout event mask
cfg_axi_compl_mask	Input	[15:0]	Individual completion event mask
cfg_axi_thresh_mask	Input	[15:0]	Individual threshold event mask
cfg_axi_perf_mask	Input	[15:0]	Individual performance event mask
cfg_axi_addr_mask	Input	[15:0]	Individual address match event mask
cfg_axi_debug_mask	Input	[15:0]	Individual debug event mask
cfg_cg_enable	Input	1	Enable clock gating
cfg_cg_idle_count	Input	[CG_IDLE_COUNT_WIDTH-1:0]	Idle cycles before gating
cfg_addr_check_enable	Input	1	
cfg_addr_range_enable	Input	[(N_ADDR_RANGES > 0 ? N_ADDR_RANGES : 1)-1:0]	
cfg_addr_filter_enable	Input	1	
cfg_addr_filter_low	Input	[AW-1:0]	

Port	Direction	Width	Description
cfg_addr_filter_high	Input	[AW-1:0]	
monbus_valid	Output	1	Monitor bus valid
monbus_ready	Input	1	Monitor bus ready
monbus_packet	Output	monitor_packet_t (128)	The monitor packet itself – the module's primary output. Valid when monbus_valid is high.
monbus_timestamp	Output	monbus_timestamp_t	Side-band sampled time for the packet on monbus_packet.
i_mon_time	Input	monbus_timestamp_t	Shared free-running timestamp, driven from the group/aggregator so every monitor stamps against one clock.
cfg_addr_range_low	Input	[AW-1:0] x N_ADDR_RANGES	Low bound of each address-range comparator; only present when N_ADDR_RANGES > 0.
cfg_addr_range_high	Input	[AW-1:0] x N_ADDR_RANGES	High bound of each address-range comparator; pairs with cfg_addr_range_l

Port	Direction	Width	Description
			ow.
busy	Output	1	
active_transactions	Output	[7:0]	Number of active transactions
error_count	Output	[15:0]	Total error count
transaction_count	Output	[31:0]	Total transaction count
cg_gating	Output	1	Gated clock is stopped
cg_idle	Output	1	No activity observed
cfg_conflict_error	Output	1	Configuration conflict detected
cfg_start_event_sel	Input	[2:0]	
cfg_end_event_sel	Input	[2:0]	
cfg_start_trigger	Input	1	
cfg_end_trigger	Input	1	
cfg_window_force_close	Input	1	
window_active	Output	1	
window_cycles	Output	[31:0]	
perf_prod_cycles	Output	[31:0]	
perf_bp_cycles	Output	[31:0]	
perf_starv_cycles	Output	[31:0]	
perf_idle_cycles	Output	[31:0]	
perf_beat_count	Output	[31:0]	
perf_byte_count	Output	[63:0]	
perf_burst_count	Output	[31:0]	

17.4 Functional Description

17.4.1 Block diagram



axil5_slave_wr_mon_cg block diagram

`axil5_slave_wr_mon` (core plus ocelus monitor) is one child box (see its own diagram), clocked by `gated_aclk` from `amba_clock_gate_ctrl`. The controller's label lists the activity terms, which add `monbus_valid` and `active_transactions` so the clock keeps running while the monitor still has packets or open transactions. Monitor-bus packets and the monitor configuration pass straight through to the child.

AXI5-Lite adds the transaction ID (`ID_WIDTH`, no enable; mandatory on this Subordinate-side module) and three optional signal groups – USER, TRACE and POISON, the ones IHI 0022L Table B2.2 allows – each gated by its own `ENABLE_*` parameter. A group contributes to the packed SKID payload only when it's enabled — turn it off and those bits simply aren't in the vector.

Group	Parameter	Signals on this module	Width
ID	`ID_WIDTH` (no enable)	`AxID`, `BID`, `RID`	`ID_WIDTH` (0..32; >= 1 on slave-side modules)
USER	`ENABLE_USER`	`AxUSER`, `WUSER`, `BUSER`, `RUSER`	`USER_WIDTH`
TRACE	`ENABLE_TRACE`	`AxTRACE`, `BTRACE`, `RTRACE`	1
POISON	`ENABLE_POISON`	`WPOISON`, `RPOISON`	one bit per 64

Group	Parameter	Signals on this module	Width
			data bits

(Only the channels this module carries are present; a read module has no W or B channel, so it has no WPOISON or BUSER.)

17.4.2 With no ID and every group disabled, this IS the AXI4-Lite module

The packed payload width of a no-ID, fully-disabled build equals its AXI4-Lite counterpart's, channel for channel:

Payload	AXI4-Lite	AXI5-Lite, `ID_WIDTH=0`, groups off	AXI5-Lite defaults (`ID_WIDTH=4`, groups on)
`ARSize`	35	35	44
`RSize`	34	34	44
`AWSize`	35	35	44
`WSize`	36	36	41
`BSize`	2	2	11

(at `AXIL_ADDR_WIDTH = AXIL_DATA_WIDTH = 32` and the default widths. `ID_WIDTH = 0` is legal only on the Manager-side `axil5_master_*` modules; a Subordinate must carry the ID, so the `axil5_slave_*` modules require `ID_WIDTH >= 1` and their minimum payload is the AXI4-Lite one plus the ID on AR/R or AW/B.)

That equivalence is what `val/amba/test_axil5_axil4_equivalence.py` pins: it runs the same AXI5-Lite BFMs, built with no ID and no groups, against both `axil4_master_rd` and `axil5_master_rd` elaborated with `ID_WIDTH=0` and every `ENABLE_*` = 0.

17.4.3 It transports; it does not interpret

The ID is carried in both directions like every other field: AxID crosses with the request, and BID/RID cross back with the response unchanged. Reflecting AxID onto BID/RID is the Subordinate's job (IHI 0022L A5.1.1); this module only guarantees that whatever ID it is given leaves on the other side. With `ID_WIDTH = 0` (Manager-side modules only) the ID outputs are driven 0 – the "tied LOW" of A5.1.1 – and the ID inputs are ignored.

TRACE and USER are carried end to end unmodified. POISON is carried – never generated, never checked. Those behaviours belong to the endpoints on either side, and nothing in this module implements them.

AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP are not carried. Table B2.2 of IHI 0022L marks them "not present" on AXI5-Lite and B2.1.5 states exclusive accesses are not supported; earlier versions of this module carried them behind `ENABLE_*` parameters (removed by issue #102).

A disabled group's output is driven to zero rather than left dangling, so an integrator who disables a group downstream of one that enables it sees a defined value instead of X.

17.5 Design Notes

A handful of traps specific to the monitored variants. Each one is written down because someone hit it.

The monitor observes the ID, not the optional groups. Since amba #104 the wrapper feeds AxID and BID/RID to `axi_monitor_filtered`, so a Subordinate that reorders responses across IDs (permitted on AXI5-Lite, IHI 0022L B2.1.5) has each response paired with its own transaction, and a completion packet carries the ID as its channel. W carries no ID and is attributed in AW order (A5.5), including W that arrives ahead of its AW (A2.6). The full monitor keys on at most 8 ID bits, so `ID_WIDTH` > 8 is refused at elaboration here (the `_monlite` wrappers take any width). There are no ports for TRACE, USER or POISON, and POISON is not validated.

`ACLK_MHZ` is not decoration. It builds the microsecond tick LUT in `counter_freq_invariant`. Leave it at the 100 MHz default on a 90 MHz part and every microsecond-denominated timeout is wrong — silently. That's the worst kind of wrong.

`NUM_BANKS` > 1 on a WRITE monitor requires `USE_WDATA_ORDER_Q = 1`. `axi_monitor_trans_mgr` fails elaboration otherwise; the error names the combination. At least that one fails loud.

A filter parameter only decides whether the logic is SYNTHESISED. A build that sets `ADDR_FILTER_ENABLE` but leaves `cfg_addr_filter_enable` low filters nothing and looks broken. The parameter and the runtime port are both required.

17.6 Timing Characteristics

17.6.1 Buffer Depths and Latency

Parameter	Default	Channel
`SKID_DEPTH_AW`	2 entries	Skid depth on the AW channel
`SKID_DEPTH_W`	2 entries	Skid depth on the W channel
`SKID_DEPTH_B`	2 entries	Skid depth on the B channel

Each channel traverses one `gaxi_skid_buffer`. That module registers both `rd_valid` and the storage array, so the **1-cycle input-to-output latency applies on every transfer, including the unstalled case** – there is no combinational bypass from the upstream payload to the downstream one. Full throughput (one transfer per cycle) is still sustained once the pipeline is primed; the depth sets how much backpressure can be absorbed before it propagates upstream, not the steady-state rate.

Legal depth range is 2..8 inclusive, odd values included.

17.6.2 Optional-group effect

The AXI5-Lite optional groups widen the packed skid payload but do not add a pipeline stage: `ARSize`, `AWSize`, `WSize`, `RSize` and `BSize` are conditional sums over the `ENABLE_*` parameters, so disabling a group narrows the storage without changing latency.

17.7 Usage Examples

Every parameter and port below is taken from the module declaration; the elisions name the remaining members of each group, all of which appear in the Ports table above. Override only what your integration needs, and never override a derived parameter – see the note under Parameters.

```
axil5_slave_wr_mon_cg #(
    .SKID_DEPTH_AW       (2),
    .SKID_DEPTH_W        (2),
    .SKID_DEPTH_B        (2),
    .AXIL_ADDR_WIDTH     (32),
    .AXIL_DATA_WIDTH     (32),
    .USE_MONITOR         (1'b1),
    .N_ADDR_RANGES       (0),
    .ADDR_FILTER_ENABLE  (1'b0),
    .ACLK_MHZ            (100),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .UNIT_ID             (8'h02),
    .AGENT_ID            (16'h0015),
    .MAX_TRANSACTIONS    (8),
    .ENABLE_FILTERING    (1),
    .ADD_PIPELINE_STAGE  (0),
    .ENABLE_ERROR_LOGIC  (1'b1),
    .ENABLE_TIMEOUT_LOGIC(1'b1),
    .ENABLE_COMPL_LOGIC  (1'b1),
    .ENABLE_THRESHOLD_LOGIC(1'b1),
    .ENABLE_PERF_LOGIC   (1'b1),
    .ENABLE_DEBUG_LOGIC  (1'b0),
    .CG_IDLE_COUNT_WIDTH (4),
```

```
    .ID_WIDTH           (4),
    .USER_WIDTH         (4),
    .ENABLE_USER        (1),
    .ENABLE_TRACE       (1),
    .ENABLE_POISON      (1)
) u_axil5_slave_wr_mon_cg (
    // clock/reset
    .aclk               (aclk),
    .aresetn            (aresetn),
    // status
    .cam_clear          (cam_clear),
    // ... `fub_axil_awaddr`, `fub_axil_awprot`, `fub_axil_awlock`,
+36 more
    // s_axil_aw
    .s_axil_awid        (s_axil_awid),
    // ... `s_axil_awaddr`, `s_axil_awprot`, `s_axil_awuser`, +3 more
    // s_axil_w
    .s_axil_wdata       (s_axil_wdata),
    // ... `s_axil_wstrb`, `s_axil_wuser`, `s_axil_wpoison`, +2 more
    // s_axil_b
    .s_axil_bid         (s_axil_bid),
    // ... `s_axil_bresp`, `s_axil_buser`, `s_axil_btrace`, +2 more
    // configuration
    .cfg_monitor_enable  (cfg_monitor_enable),
    // ... `cfg_error_enable`, `cfg_timeout_enable`,
`cfg_perf_enable`, +30 more
    // monitor bus
    .monbus_valid       (monbus_valid)
    // ... `monbus_ready`, `monbus_packet`, `monbus_timestamp`
);
```

17.8 Related Modules

- axil5_master_rd
- axil5_master_rd_cg
- axil5_master_rd_mon
- axil5_master_rd_mon_cg
- axil5_master_wr
- axil5_master_wr_cg
- `axil4_slave_wr_mon_cg` – the AXI4-Lite counterpart
- AXI4-Lite modules

17.9 Testing

`val/amba/test_axil5_slave_wr_mon_cg.py` drives this module with the AXI5-Lite BFMs and **the ID and every optional group enabled** — `TBClasses/axil5` sets the group widths in `COMPONENT_KWARGS` to mirror the RTL defaults. A BFM configured differently from its DUT is a bind failure, which is the loud version of the mistake — and the version you'd rather get.

Every transaction the Manager-side BFM issues carries a rotating non-zero ID and must come back with the same BID/RID (`bin/TBClasses/axil5/axil5_id_check.py`); the inherited AXI4-Lite phases alone would only ever send ID 0, which a module that dropped the ID would pass.

The testbench class is the AXI4-Lite one with the component factories swapped, so every phase, check and randomizer sweep has a single definition, and a fix to the AXI4-Lite flow reaches this module automatically. That's exactly how you want a variant family to work.

17.10 References

- Arm. *AMBA AXI Protocol Specification*, ARM IHI 0022, Issue L, Aug. 2025. developer.arm.com/documentation/ihi0022/l – §B2.1.5 "AXI5-Lite" and Table B2.2 signal matrix; carried groups per §A12.5 User defined signaling, §A12.3 Trace signals, §A5.1.1 transaction ID signals, §A16.1 Poison; handshake on the AW, W and B channels per §A2.3.1/§A2.3.2; register stages per §A2.5. Table B2.2 marks AxLOCK, AxMPAM, AxMECID, AxNSAID and xLOOP not present on AXI5-Lite and AxID/BID/RID mandatory on a Subordinate (YS): this module carries the ID (`ID_WIDTH`) and none of the not-present signals (issue #102). Data width per Table A3.19 (any AXI DATA_WIDTH; Table B2.4 adds no AXI5-Lite constraint).
- Arm. *AMBA AXI and ACE Protocol Specification*, ARM IHI 0022, Issue H, 2020. developer.arm.com/documentation/ihi0022/h – Part C ch. C2 "AMBA AXI5-Lite", §C2.2 comparison with AXI4-Lite and §C2.5/§C2.6 upgrading an AXI4-Lite master/slave: the `ID_WIDTH=0`, every-group-disabled Manager-side build is the AXI4-Lite-equivalent port.
- Benini, L., Bogliolo, A., De Micheli, G. "A Survey of Design Techniques for System-Level Dynamic Power Management." *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – fixed-timeout shutdown: gate after `cfg_cg_idle_count` idle cycles, wake on any VALID.
- Benini, L., De Micheli, G. "Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines." *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based glitch-free clock gate under `clock_gate_ctrl`.

- READY held LOW while gated is legal: a receiver may keep READY LOW until it chooses to accept (IHI 0022H §A3.2.1; IHI 0022L §A2.3.1). No paper (standard technique).
- The monitor itself (`axi_monitor_filtered`) is referenced in the monitor pages; on this port it sees single-beat transactions carrying the AXI5-Lite ID (IHI 0022L B2.1.5; W in AW order, A5.5).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/amba/references/README.md`; the clock-gating papers are archived in `rtl/common/references/` (local PDF: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

17.11 Navigation

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